

Prague University of Economics and Business

Faculty of Informatics and Statistics



THE ECONOMETRIC ANALYSIS OF CROWDFUNDING PROJECTS

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Author: Maria Papush

Supervisor: Mgr. Vladimír Holý, Ph.D.

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P r o h l á š e n í

Prohlašuji, že jsem bakalářskou práci na téma
„Econometric analysis of crowdfunding campaigns“
vypracovala samostatně s využitím literatury a informací,
na něž odkazuji.

V Praze dne 30. června 2023

Maria Papush

Obsah

Introduction	5
1 Crowdfunding Platforms	6
2 Theoretical Part	11
3 Practical Part	15
TEST AND TREND DATASET	Error! Bookmark not defined.
Discussion	35
Conclusion	36
List of references	37
1. Appendix – Insight into selected crowdfunding platforms	39
1.1. Market size of crowdfunding worldwide in 2021 with a forecast for 2028	51
1.2. Most successfully completed Kickstarter projects as of May 2022, by amount of total funds raised	52
1.3. Overview of projects and USD turnover on crowdfunding platform Kickstarter as of November 2022	53
1.4. Number of successfully funded projects on Kickstarter as of November 2022, by amount of money raised	54
1.5. Distribution of unsuccessfully funded projects on crowdfunding platform Kickstarter as of November 2022, by share of funding reached	55
1.6. Largest IndieGoGo campaigns worldwide as of March 2022, by amount raised	56

Abstract

Crowdfunding is one of the most interesting and inspiring topics of recent times and is only getting more popular as time goes by. Studying it allows you to better understand the incentive that arouse public interest, make assumptions about how information is transmitted and wonder as far is possible to predict the behavior of the crowd.

Not surprisingly, there are more and more studies exploring the subject.

This paper provides a brief overview of articles on the topic of crowdfunding, mainly in terms of the factors that affect the success of the project. In the practical part of the current bachelor thesis, there is a study of the key success factors of the project based on the data of the two most famous reward-based crowdfunding platforms Kickstarter and IndieGoGo. The practical part of this thesis is to build the Lenear Regression and Logistic Regression, testing the quality of the models and eliminated the key factors of a project success.

Keywords

Crowdfunding, linear regression, logistic regression, review, diagnostic, key factors

Introduction

Between relatively young, but fast-growing ways of gaining capital, both for individuals and legal entities, belong so-called crowdfunding. The rapidly and significantly growing markets for crowdfunding, in which individuals directly finance other individuals or companies without financial intermediation, are applied to various types of financing worldwide and provide a significant potential pool of capital (Hildebrand, Puri, Rocholl, 2013). Mallek 2014 This method is based on gaining capital for certain project or entrepreneur activities through the public. Crowdfunding allows companies and individuals to obtain financing in an alternative and more effective way than traditional sources (Borrero-Domínguez, Cordon-Lagares, Hernández-Garrido, 2019). Individual members of the public, or the crowd, contribute by smaller amounts of financial means for realization of certain project in a given timeframe. This newly emerged crowd can be technically considered as a group-creditor. An environment where the financing part in a crowdfunding model takes place is an Internet. Internet basically serves as a connecting bridge between the creditors (investors) and entrepreneurs seeking for financial resources. Chinese study (2019) indicates high importance of an online communication, especially at the early stage. An entrepreneur's idea is typically being promoted on the same crowdfunding platform, and, if agreed amount of money is raised, an entrepreneur may start realizing the project. Otherwise, if the finance goal (a certain money cap) is not met, the finances invested are returned back to the creditors.

However, from an entrepreneurial perspective, even if the crowdfunding campaign fails to raise the funding target, the media coverage, as stated in a recent study (2021), may lead to a number of significant business avenues, which ultimately help launching the product onto the marketplace (Wang, Johnston, Caraher, 2021). In relation to such a scenario, crowdfunding campaign then serves as an instrument of an initial market validation, collecting innovative ideas from several perspectives (an entrepreneurial, an investor's, a platform runner's), and, of course, receiving valuable feedback.

1 Crowdfunding Platforms

1.1 History

The history of crowdfunding can be traced back to 1800s. Modern forms of crowdfunding, directly relevant to the topic of this thesis, can be first historically spotted, most probably, in 2000, in the United States, when ArtistShare was launched as a website where musicians could seek donations from their fans to produce digital recordings (Harris, Lam, Zhao, 2019). Later, it has evolved into a fundraising platform for music, film/video, and photography projects. Today, market size of crowdfunding worldwide is estimated at approximately USD 13.64 billion and is expected to grow up to USD 28.92 billion in 2028 (Statista, 2022). For more numbering figures please refer to the appendix part at the end of the current thesis.

1.1 Types

There are four basic types of crowdfunding methods to discuss in this thesis: (i) reward-based model, (ii) donation-based model, (iii) peer-to-peer lending and (iv) equity-based crowdfunding. In reward-based crowdfunding (RBC) model investors may receive certain rewards, bonuses, company's products or gifts (examples are IndieGoGo, Kickstarter). Of course, more appealing model, from an investing point of view, is based on acquiring a certain share of the company, i.e., becoming a co-owner – equity-based crowdfunding in a nutshell, which rocketed especially in the beginning-mid 2010's. As a less appealing model would appear so-called donation-based crowdfunding (DBC), when, from the economic point of view, investors behave irrationally and do not expect any material reward from their donation (JustGiving). DBC is similar to charities and contributes mostly to social projects or local initiatives. Reward-based crowdfunding (RBC) will be discussed the most in the following parts of this thesis.

An important part of crowdfunding as a capital-gaining method is the fact, that young companies, or start-ups, do not have to guarantee the project with its own assets, which would be the banking loan case. As such, crowdfunding plays a role in filling the funding gap traditionally identified for small businesses (Hervé, Schwienbacher, 2018). The other advantage of the crowdfunding method is a chance to get the first customers even before launching the core business activity. The customers, in their turn, can provide valuable feedback for a newly creating product or service, and therefore to participate in the innovation process along with the founders.

In a relatively recent study (2018) it was confirmed, that pledging conditions, project-related aspects, as well as founder-related characteristics influence funding success. Furthermore, risk disclosure and emotional appeal have positive effects on funding success (Koch, Siering, 2018). Nowadays, sophisticated machine learning algorithms are used to predict whether the project will reach its funding goal or not.

1.2 Regulation

Speaking about whatever investment instrument, local regulatory approaches have to be taken into consideration – and that applies to both perspectives, an entrepreneur's and the consumers. Different regulatory environments, as well as country economic structures, may have significant impact on the investor protection, innovation process, popularity of the crowdfunding platforms themselves, and, therefore, the projects presented. An investor protection, specifically, may even harm small firms and entrepreneurial initiatives, if it is too strong. Securities regulation primarily deals with regulating large issuances, and, therefore, impose significant compliance costs that are prohibitively high for small firms (Hornuf, Schwienbacher, 2016). On the other hand, given the size of RBC and its importance to individuals and small businesses, the lack of rules and regulations highlights the burning issues of potential fraud, either intentional or unintentional (Harris, Lam, Zhao, 2019).

1.3 Implications

The implications of using crowdfunding platforms are richer than can be assumed by seeing them simply as an alternative funding source to reduce financial expenses (Luo et al., 2022). Recent study (2022) discussed, in a context of microfinance institutions (MFIs), that the effect of partnership with online crowdfunding is not only a short-term remedy that provides access to low-cost funds, but is also a trigger to induce long-term structural changes.

The authors also discuss whether the crowdfunding should be encouraged on a government level. Bringing up the regulatory framework for consumer lending, there is evidence from a clean experiment (2013), that shows that proper incentives are crucial for giving borrowers access to credit and to induce lenders to carefully screen loan applicants. That matter will not be developed in this bachelor thesis any further – complex political, economic, and social factors should be considered for the specific jurisdiction (as authors indicate raising the subject of macroeconomic indexes such as liquidity indicators, again, more relevant in MFIs context).

1.4 Reward-Based Crow

Kickstarter, the world's largest reward-based crowdfunding website, uses an all-or-nothing funding model: the creator will receive the funding committed by backers and the backers will obtain their rewards if and only if a project is successfully funded, i.e., reaches its funding goal by the end of its campaign (Dai, Zhang, 2019). In this model, setting the funding goal is a crucial criterium. There are basically two main approaches: setting the funding goal slightly below or slightly above the actual capital cap set in the business plan. Study (2019) shows that funding accumulates faster when projects are slightly below their funding goals. Prosocial motives are entering significantly and therefore must be considered along with the machine learning methods when modifying and predicting the crowdfunding campaign outcome.

1.5 Pirsing strategy

Crowdfunding platforms adopt a pricing approach for all participants involving fixed transaction rates (Tang, Lu, Huang, Liu, 2021). Pricing strategies of the crowdfunding platforms can be generally divided into two groups: symmetric, i.e., charging both investors and the entrepreneurs, or asymmetric, meaning charging one party only (typically leaving investors free of charge, but charging project initiators instead). Minimum investment amount strategy can be applied for any symmetric and asymmetric group. However, minimum investment amount may discourage some investors because of their limited funding sources.

1.6 Covid links

Despite the COVID-19 worldwide pandemic, the world's most popular RBC-based platform, Kickstarter, states the quantitative number of visitors remained the same in 2020, funding success rate for projects has risen to 51 percent, outperforming the average from September 2019 by five percentage points (Kickstarter.com, 2020). According to the authors there was just about 4-5 months cooldown in the number of live projects; then, activity started to recover. Kickstarter initiated innovative activities through Inside Voices specifically for the quarantine ideas. In parallel, similar platform was launched for cultural institutions – Lights On, which helped running venues of all kinds. Primary consideration when launching the project, in the light of pandemic, have become people's health and wellbeing.

Similar observations were confirmed in this year's study from Europe, Netherlands. The number of online donations via crowdfunding did not show any significant change during the COVID-19 pandemic (Hasanefendic, Teunenbroek, 2022). In this regard, the authors underline an importance of an off-line events, i.e., off-line marketing activities, to attract the potential investors. However, study discovered slight qualitative increase in the average donation amount (by 14%, from EUR 43 to EUR 49).

There has been conducted another research in relations to the COVID-19 worldwide pandemic. An unusual case of collective fundraising was investigated in the UK (2022), specifically in a healthcare-related crowdfunding area. GBP 150 million were total raised as an urgent appeal through charitable donations. The appeal was studiously apolitical, with COVID-19 clearly identified as an impersonal natural disaster that justified widespread fundraising (Stewart, Nonhebel, Moeller, Bassett, 2022). Appeals were dominated by themes of solidarity and duty, which, in turn, mobilized nationalistic narratives. Such a phenomenon should be understood within the pandemic timeframe, in a health system context.

Another study paper from Canada (2022) uniquely situates attitudes toward COVID-19 vaccines in the context of financial need (Snyder, Goldenberg, Crooks, Katz, 2022). 781 US campaigns (with 5 in multiple categories, see further) indicated on Facebook were divided into three categories: utopian, cautious and sceptic. However, regardless the campaign runners vision (either assuming the COVID vaccine is an unconditional mean to win the pandemic, or, the other way around, viewing vaccine as a fundamentally flawed approach), the result of the study is the fact that crowdfunding priorities reflect public policy priorities – meaning, how the crowdfunding preferences in choosing the particular campaign change over time accordingly to the public values distribution.

1.7 Other relevant literature review

Entrepreneurs are eager to know the prospects of their campaigns; investors also want to fund promising campaigns with high quality and low uncertainty (Bao et al., 2022). The following area represents of a great level of interest for both involved parties, therefore, studies on success determinators and success prediction were inducted in the past years quite widely. In the following bachelor thesis I will review relevant aspects, especially for the purposes of a practical part, from earlier studied success factors, up to the most recent prediction researches.

Ethan Mollick, one of the pioneers of crowdfunding research, in his 2013 paper study based on a dataset (extracted from Kickstarter) of over 48,500 projects with combined funding over USD 237 million, discovered a list of common features, which influence successful fundraising significant enough to be mandatory considered. Those features are basically a set of quality signals (which can be considered as a crowdfunding campaign success factors): fundraising duration, funding level, Kickstarter's promotion, project videos available, regular and rapid updates, spelling in project pitches, entrepreneur's network size, geographic constraints (across the US), the complexity of the project and the project timing. According to the author's conclusion, for funders, careful planning is required to set the appropriate goals. More specific is the project updates analysis conducted in 2014, which discovered a significant finding regarding the investor and project runners' communication: how project creators communicate with potential funders during a campaign is more predictive of success than the representation of the project page (Xu, Yang, Rao, Fu, Huang, Bailey, 2014).

In further research some other significant variables were found. According to RBC analysis conducted in Spain (Borrero-Domínguez, Cordon-Lagares, Hernández-Garrido, 2019), the following dependent variables are statistically significant for crowdfunding project success: location, experience, human capital and gender. However, the quality of the dataset of this research has to be underlined (hand-collected data from the Spanish Goteo platform), as the authors state in the limitations part of their research.

A study specifically on a relation between project description and crowdfunding success was conducted using data collected from Kickstarter (Zhou, Lu, Fan, Wang, 2016). Authors identified three exemplary antecedents from the language use of project description: length, readability, and tone. Since newly introducing products and services are often completely unknown for the funder, long and enough detailed description of the project is strongly recommended. Although too much information can negatively affect – excessive information could reduce readability and increase psychological fatigue. The potential fund-providers could be confused by the variability of the projects to fund and avoid dealing with this stressful situation. Thus, the information provided to potential fund-providers during a crowdfunding campaign has a curvilinear (inverse-U shaped) relationship with crowdfunding success (Thapa, 2020). According to the paper the funding success was maximum amount crowdfunding campaigns that used 2000–3000 words.

The tone will, in its turn, show positive and patient attitude, which helps the potential investor to emphasize the project even emotionally.

The other quality signals (speaking in terms of Mollick) are past expertise and past experience of project owners. However, the authors identified the factors related to the project descriptions as more significant and having more practical impact on the crowdfunding project success.

In the most recent study (2022) applicable to RBC was stated that Kickstarter, IndieGoGo and similar platforms (with an open comments section) can predict the outcomes of projects by monitoring comments (Bao et al., 2022). The authors identify the comments as a beneficial instrument for every party – which stands in line with the previous research on communication, as a part of a crowdfunding campaign. Comment quantity had been confirmed as an important factor in another study (Jiang et al., 2020), and it was confirmed as significantly positive, whereas reward execution moderates the impact of goal. Project comments and project description is mostly an entrepreneur's perspective, and those have to have positive sentiment to achieve a positive impact on crowdfunding performance (while a negative one would have a negative impact).

In a relatively recent study paper (Li et al., 2021) the importance of video clip available on project's page was discussed. At Kickstarter preparatory material for project initiators, Kickstarter suggests that the key to demonstrating preparation is to include a video: "Rule #1 for Kickstarter videos: make one! There are few things more important to a quality Kickstarter project than video. Skipping this step will do a serious disservice to your project".

The results shows that the "entrepreneur could improve the funding outcomes by gazing less, appearing early, and reducing speech hesitations in a pitch video". The quality of the video was also important, so was the music length. Smiling come out surprisingly nonsignificant.

Crowdfund smart, not hard – Understanding the role of online funding communities in crowdfunding success (Hassna, 2022) is a one of the most popular papers about backers' behavior. The author begins with the statement than the large amount of pledged money came from the online organization who identify with the campaign and share same interests and ideologies. Congruence between the crowdfunding campaign and targeted online funding communities is associated with higher success rate.

2 Theoretical Part

2.1 Linear regression

For modeling the relationship between a dependent variable and independent variables has been used the Linear Regression model. For fitting a regression line was chosen the method of least-squares. Other possible options for constructing a model will be discussed further in the "Discussion" section.

The regression function is formulated:

$$y_i = \beta_0 + \beta_1 x_i + \dots + \beta_m x_{im} + \varepsilon_i = \mathbf{X}_i^T \boldsymbol{\beta} + u_i,$$

$$i = 1, \dots, n$$

or

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \text{ where}$$

$\mathbf{y}$ vector of dependent value,

$\mathbf{X}$ matrix of independent variables,

$\boldsymbol{\beta}$ are regression parameters,

$\mathbf{u}$ is vector of error term.

This method calculates the best-fitting line for the observed data by minimizing the sum of the squares of the vertical deviations from each data point to the line.

Linear regression assumptions

The Gauss Markov theorem tells that if a certain set of assumptions are met, the ordinary least squares estimate for regression coefficients gives the best linear unbiased estimate (BLUE).

1. Linearity: the parameters we are estimating using the OLS method must be linear.
2. Random: our data must have been randomly sampled from the population. The observations should not come from repeated measurements or matched data.
3. Non- Multicollinearity: the regressors being calculated aren't perfectly correlated with each other.
4. Exogeneity: the regressors aren't correlated with the error term.

$$E(u|X) = 0$$

5. Homoscedasticity: no matter what the values of our regressors might be, the error of the variance is constant. Non – autocorrelation, the residual are independent on each other.

$$\text{var}(u|X) = \sigma^2 \mathbf{I}, \sigma^2 \in (0, \infty)$$

6. There is also 6th assumption: the normality distribution of residuals. The OLS estimator will be BLUE if the first five conditions are met. The sixth is necessary for most standard tests to work correctly.

$$\text{var } u|X \sim N(0, \sigma^2 \mathbf{I},)$$

This is a soft assumption, this requirement is not necessary should be met for models with a big sample size datasets.

2.2 Logistic regression

The beta parameter, or coefficient, in this model is commonly estimated via maximum likelihood estimation (MLE).

Logistic function is formulated:

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

The logit (log odds) function is defined as the inverse of the standard logistic function:

$$\begin{aligned} \text{logit}(E[Y_i | x_{1,i}, \dots, x_{m,i}]) &= \text{logit}(p_i) = \ln\left(\frac{p_i}{1 - p_i}\right) \\ &= \beta_0 + \beta_1 x_{1,i} + \dots + \beta_m x_{m,i} = \boldsymbol{\beta X} \end{aligned}$$

The beta parameter, or coefficient, in this model is commonly estimated via maximum likelihood estimation (MLE).

Logistic regression assumptions

Logistic regression does not make many of the key assumptions of linear regression. Logistic regression does not require a linear relationship between the dependent and independent variables, the residuals do not need to be normally distributed, heteroscedasticity is automatically assumed to exist.

1. The dependent variable is binary
2. Observations should be independent of each other.
3. No perfect multicollinearity among the independent variables.
4. Linearity of independent variables and log odds.
5. A large sample size. A general guideline is at minimum of 10 cases with the least frequent outcome for each independent variable in a model.

3 Practical Part

3.1. Data preparation

The study is based on the data about projects launched and funded on the two biggest crowdfunding platforms, Kickstarter and IndieGoGo.

The datasets I used are available at the web site of the web robot, that collected data about Kickstarter and IndieGoGo projects from 2009 for Kickstarter and from 2010 for IndieGoGo to present time. The reference to the data source is below (datasets 1.1.2022).

<https://webrobots.io/kickstarter-datasets/>

<https://webrobots.io/IndieGoGo-dataset/>

In the Kickstarter dataset more variables are available. Moreover, the categories of the Kickstarter projects are slightly different from the categories of the IndieGoGo dataset. Thus, I decided to analyze the Kickstarter and IndieGoGo projects separately.

Kickstarter's project data were download in 9 CSV files and has been merged into the one for 29286 observations on 39 variables. There are 68125 observations and 24 variables in the IndieGoGo dataset.

For data preparation and modelling was used the R Studio program.

Kickstarter data preparation

Original dataset collects 39 variables .There is the information about count of backers, country of projects, projects category, pledged amount, goal, that the founders are trying to achieve, and the date of a deadline. Variables launched_at, created_at, state_changet_obtain the day, month and year then the project was funded.

All date variables are in UNIXTIME format, so I used the function as.POSIX to transform it to the date format.

.I have created the new column named „length“ with a duration of the project, calculated from the date then the project was launched and a deadline. I have also created columns named „launched_day “, „launched_month“ and „launched_year“ in purpose to test if this information could affect the project's final state.

The value of the variable category is not specified as a specific category name but was created as a combination of project identification numbers, source references, project names and some additional information. Thus, although they contained a set of basic categories, each row is unique. It was not even possible to separate this column into subsections because of the format used. For further testing was required to transform these values into individual category names.

Regarding to Kickstarter page, there are projects of 12 categories: Comics, Games, Food, Theater, Technology, Publishing, Photography, Journalism, Music, Film & Video, Fashion, Design, Dance, Crafts and Art. Based on this, I used function **grepl** to detect the names of the categories in text in „Category“ column and **ifelse** to replace it by the name of the category.

IndieGoGo data preparation

The data of the IndieGoGo projects was in 3 different csv files, I have merged them to the new large dataset using **rbind** function.

The dataset obtained the NA of two different types: NA and „null“. The „null“ appears mostly at the date of the beginning and the date of the closing projects. I detected the „null“ observation, replaced it by NA and then deleted them all together.

Then I created the following new columns for the further testing:

Column duration (in days), columns created_day (beginning on Sunday), created_month and created_year. Column „goal“, with the information about the aim that founder tries to archive. I calculate the values by dividing rised_amount by rised_percent.

The pledged amount of the projects in the initial dataset is given in the recipient's currency. Unfortunately, the file does not include the tax rate, so I had to find the current data and calculate the amount collected in US dollars.

For the logistic regression the Kickstarter dataset has been extended with an additional column state edit, containing 0 in case the project has reached the amount set by the founder in the target column, and zero if the target amount has not been reached. Also for further calculations for cases where the target amount was zero, the target amounts were replaced by 0.00000001. It should also be mentioned that due to difficulties in collecting dates in the original data file of the IndieGoGo platform, the minimum value for project duration was negative, minus one, for 4,000 cases. The durations for these cases were changed from -1 to 0.

Kickstarter	IndieGoGo
Duration	Duration
Created Day	Created Day
Created Month	Created Month
Created Year	Created Year
Goal	Sate edit
Converted pledged	Converted pledged Rate

Tab. 1 – added columns

(Source: own study)

3.2 Model formulation

Linear regression on the Kickstarter dataset

$$\log(\text{pledged amount}) = \beta_0 + \mathbf{K} + \mathbf{L} + \beta_2 \log(\text{goal}) + \beta_3 \text{stuff} + \beta_4 \log(\text{duration}) + \mathbf{M} + \mathbf{N} + \mathbf{T} + \beta_5 \text{spotlight} + u,$$

$$\mathbf{K}_i = \sum \mathbf{K} x_i,$$

$$\mathbf{L}_i = \sum \mathbf{L} x_i,$$

$$\mathbf{M}_i = \sum \mathbf{M} x_i,$$

$$\mathbf{N}_i = \sum \mathbf{N} x_i, \mathbf{N}_i$$

$$\mathbf{T}_i = \sum \mathbf{T} x_i,$$

where $\mathbf{K}_i$ is a category,

$\mathbf{L}_i$ is a currency,

$\mathbf{M}_i$ is created day,

$\mathbf{N}_i$ is created month,

$\mathbf{T}_i$ is created year.

Linear regression on the IndieGoGo dataset

$$\log(\text{pledged amount}) = \beta_0 + \mathbf{K} + \mathbf{L} + \beta_1 \log(\text{goal}) + \beta_2 \log(\text{duration}) + \mathbf{M} + \mathbf{N} + \mathbf{T} + \beta_3 \text{indemand} + u,$$

Logistic regression on the Kickstarter dataset

$$\text{logit}(\text{state}) = \beta_0 + \mathbf{K} + \mathbf{L} + \beta_2 \log(\text{goal}) + \beta_3 \text{stuff} + \beta_4 \log(\text{duration}) + \mathbf{M} + \mathbf{N} + \mathbf{T} + \beta_5 \text{spotlight} + u,$$

Logistic regression on the IndieGoGo dataset

$$\text{logit}(\text{state}) = \beta_0 + \mathbf{K}_i + \mathbf{L} + \beta_1 \log(\text{goal}) + \beta_2 \log(\text{duration}) + \mathbf{M} + \mathbf{N} + \mathbf{T} + \beta_3 \text{indemand} + u.$$

Pledged amount is the amount of money received in total, category – one of 12 categories, goal – money amount set by the founder, staff - staff shows whether there is a picture of the product

on the project page, duration shows the duration of the company in days, day is the day of the week the project has been started, month- the month there the projects has been started, spotlight shows the type of Kickstarter company promotion. The variable starrable was excluded from the final model due to statistical insignificance (p value – 0.93). The list of the categories could be get from the Estimated coefficients table at Appendix section.

3.3 Linear regression model testing

3.3.1 Variables significance

Almost all variables in the final model appear to be statistically significant at the 0.05 significance level, except for individual category and *currency* variables, which are actually subsets of the „factor“- type variables. None of the variables in the subgroup created month were statistically significant for model number 2 - linear regression on the IndieGoGo dataset.

In this case, the F-test of Overall Significance is used to identify the impact of the group of the variables.

The null hypothesis for the test was:

$$H^0: \sum M_i = 0,$$

$H^1: non H^0$,

M_i – regressor for the i created month.

The p-value for category is 2.2e-16 with test statistic 126.59, the null hypothesis was rejected, the variable category is significant at the 0.05 level. The factor variables created day and created month was tasted, and found to be overall significant, the null hypothesis were rejected with p-value very close to 0.

The variables starrable and spotlight representing the type of the Kickstarter promotion company were significant only for the the first model, this promotion types help projects raise more money on average, but has no impact on the success ikelihood.

It seems that projects with a high target amount raise more money on average on a Kickstarter, but they are not necessarily successful. At the same time it is the opposite for Indiegogo, the target amount does not affect the amount of money raised, but helps the project to be successful.

The duration wasnt important for the projects on the Kickstarter platform, but affect negatively the pledged amount and the success ratio on IndieGoGo projects.

The list of the significant variables is at the appendix section.

3.3.2 Model quality

For a visual analysis of the model I use The Residuals vs Fitted plot, of residuals on the y axis and fitted values on the x axis. It is basically used to detect non-linearity, unequal error variances, and outliers.

Next plot I use is the Q–Q plot (quantile-quantile plot) (**Error! Reference source not found.**), that compare the distribution of the residuals with normal distribution. On the x-axis are the theoretical quantiles of a standard normal, on the y-axis is the empirical distribution function of the standardized residuals.

The distribution of the residuals is far away from the normal, the graphs show that the model chosen (the simple regression model with the linear dependent and independent variables) does not fit the data well, so a model transformation is being required. The Adjusted R squared for the Kickstarter model was 0.048, 0.0485 for the IndieGoGo model.

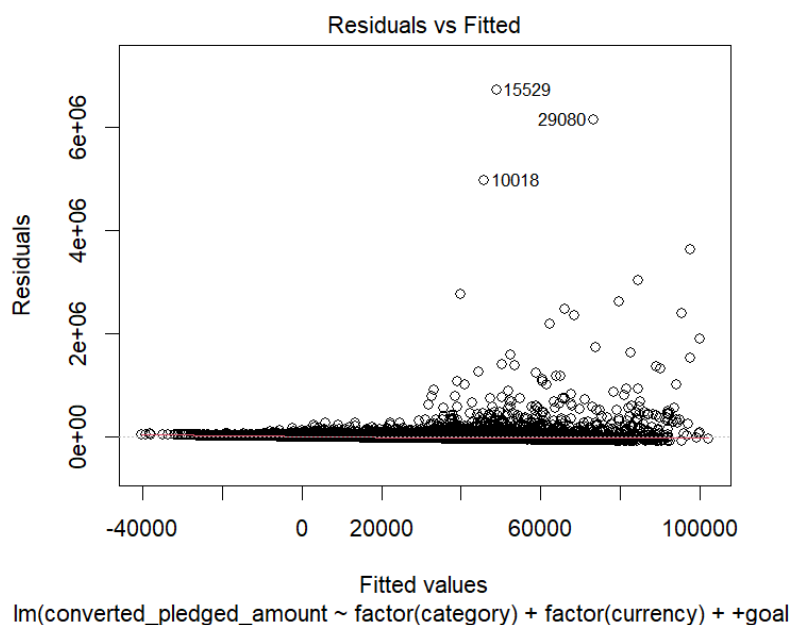


Figure 1 – Residual plot of the Kickstarter Linear model before the model transformation.
(Source: own study)

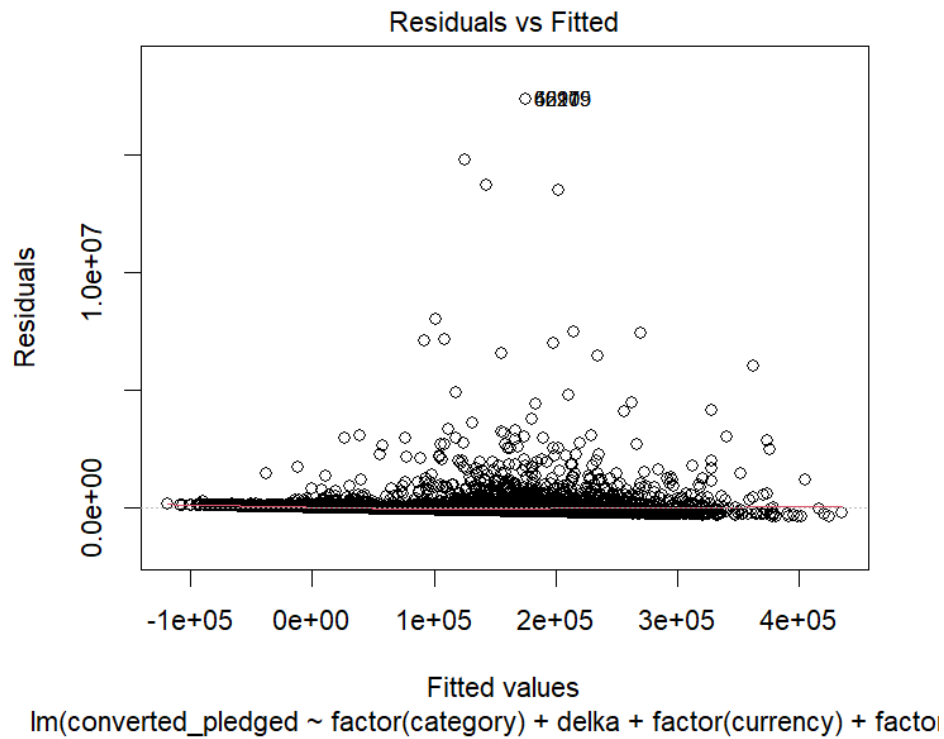


Fig. 2 – Residual plot of the IndieGoGo Linear model before the model transformation
(Source: own study)

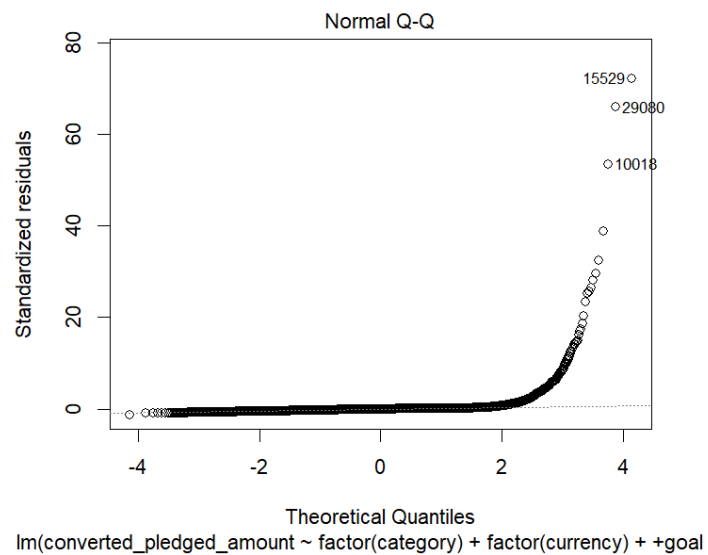


Fig. 3 – Quantile-Quantil plot of the Kickstarter Linear model before the model transformation.
(Source: own study)

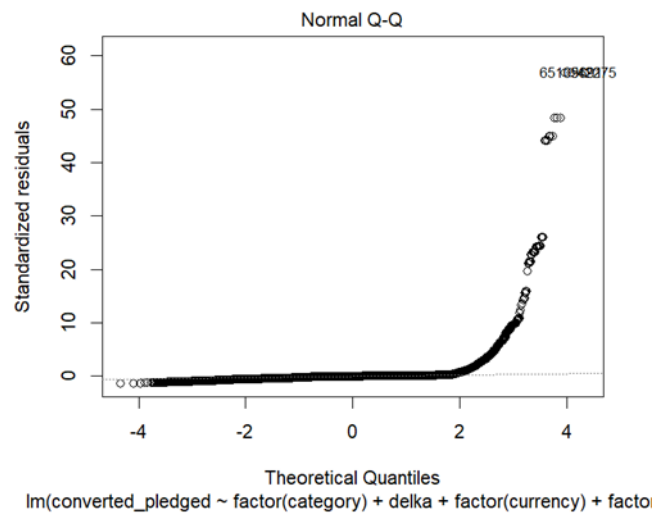


Fig. 4 – Quantile-Quantil plot of the IndieGoGo Linear model before the model transformation
(Source: own study)

3.3.3 Heteroskedasticity test

Heteroscedasticity is visible on the Residual Fitted plot, a visual impression is confirmed by the Breusch-Pagan test - the null hypotheses was rejected at 0.5 with p-value

$< 2.2e-16$, heteroscedasticity occurs in the dataset.

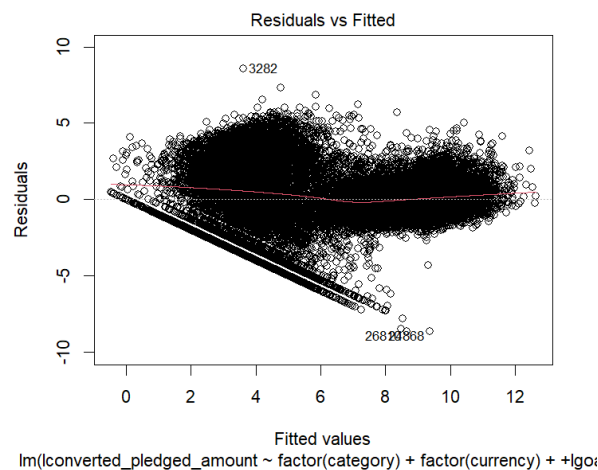
In case of the heteroscedasticity, the estimated coefficients in the model are still unbiased but the standard errors do not reflect the actual errors, and t-test values appear to be overly optimistic. The errors for the coefficients are produced smaller than they actually are.

One method of reducing heteroscedasticity is to transform the dependent variable, in this study a log transformation was chosen. The dependent variable converted pledged amount and the independent variables goal and duration was transformed to their logarithmic version.

3.3.4 Transformed model

After transforming the model, we can see that heteroscedasticity is less prominent, but still present. Therefore, in order to avoid errors in estimating the significance of the variables, robust standard errors are used.

The density plot shows the distribution of the residuals for all observations. The density plots and the Q-Q plots below show, that the distribution of residuals is closer to normal (but still not normal), and the models fit is better in general. Adjusted R squared for the Kickstarter transform model is 0.67 after the transformation, 0.558 for the IndieGoGo.



*Fig. 5 – Residual plot of the Kickstarter Linear model after the model transformation
(Source: own study)*

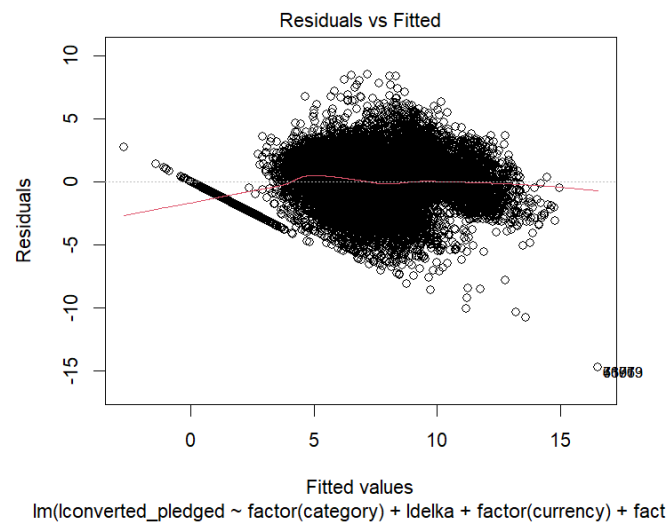


Fig. 6 – Residual plot of the IndieGoGo Linear model after the model transformation
(Source: own study)

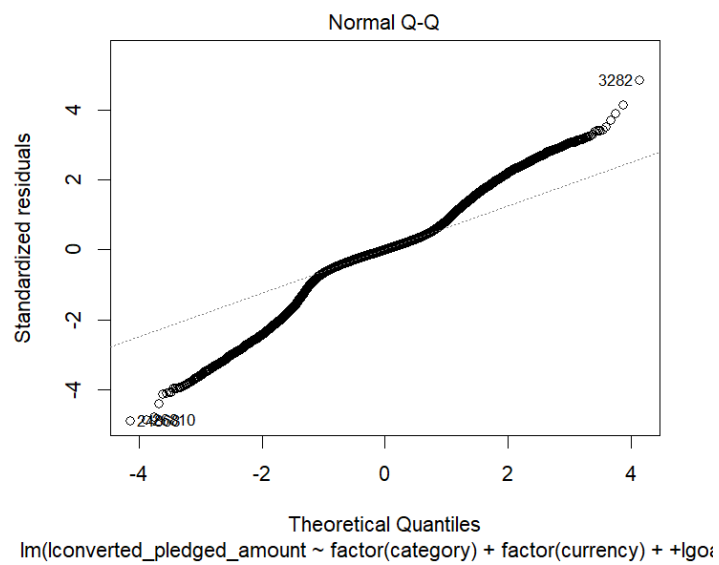


Fig. 7 – Quantile-Quantil plot of the Kickstarter Linear model after the model transformation
(Source: own study)

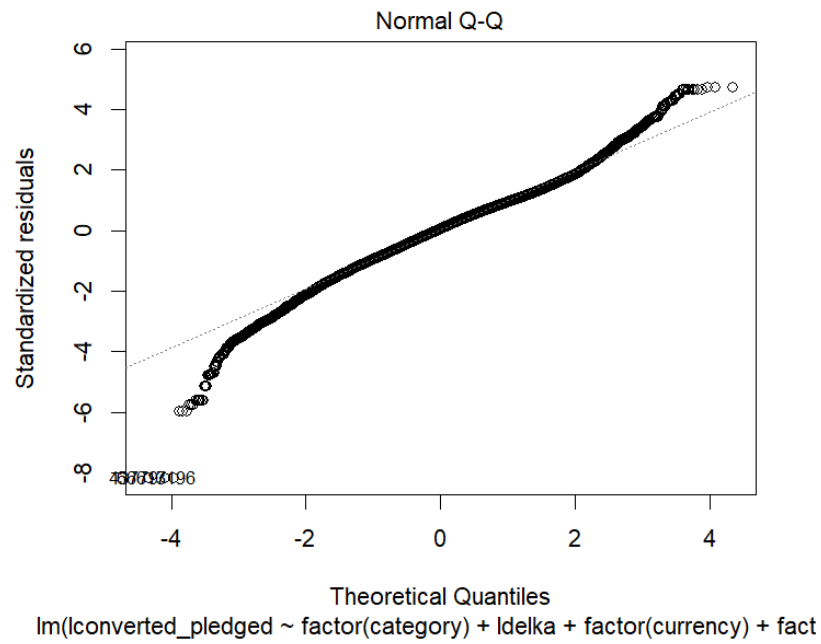


Fig. 8 – Quantile-Quantil plot of the IndieGoGo Linear model after the model transformation
(Source: own study)

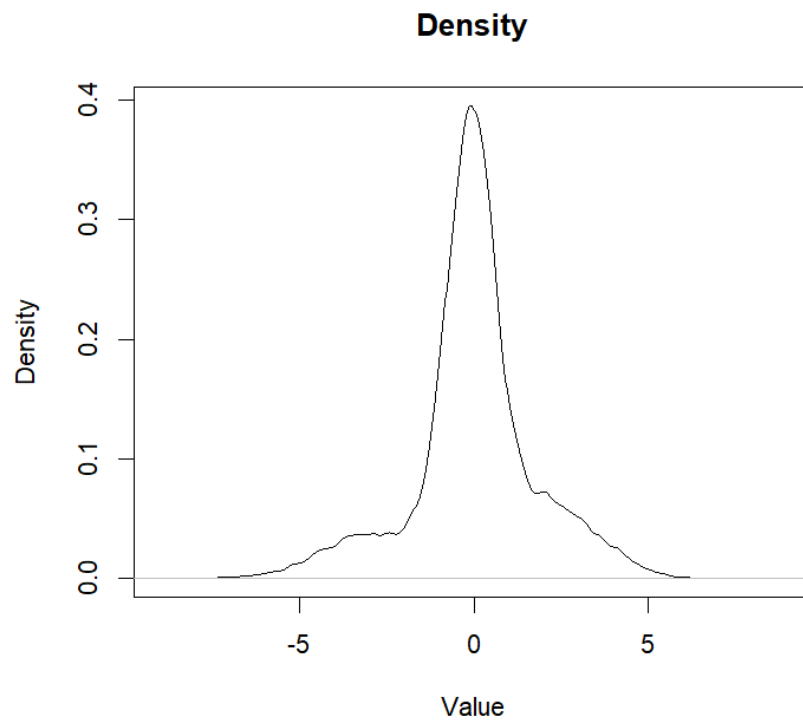
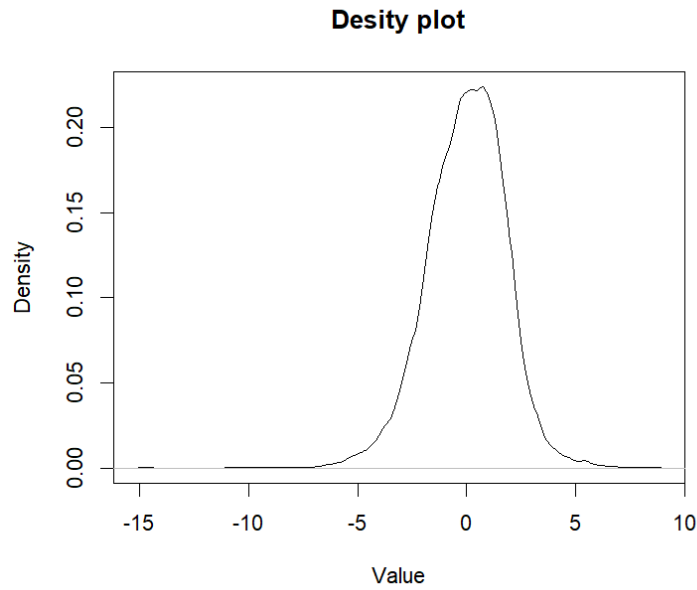


Fig. 9 – Density plot of Kickstarter model
(Source: own study)



*Fig. 10 – Density plot of IndieGoGo model
(Source: own study)*

3.4.5 Outliers

The plots for the new models above show that the both Kickstarter and IndieGoGo datasets contain outliers that appear to be organized in a linear pattern. These outliers is the observations with log (converted pledged amount) values equal to 0, or converted pledged amount equal to 1.

These variables are transformed variables, in a logistic transformation the variables were transformed according to the formula $lconverted\ amount_i = \ln(1 + converted\ amount_i)$.

Nevertheless, as the formula has been correctly formulated, it is possible there is measurement error. This could be an important issue to investigate in further studies based on those datasets.

Regardless of this, the formulated transformed models still give the impression of being the best possible model transformation, at least for the analysis methods that has been stated in this study .

The graphs below represent the distribution of converted pledged amount and the distribution of log (converted pledged amount). Another outliers are shown at the plots below. Outliers elimination was based on the Cook's distance.

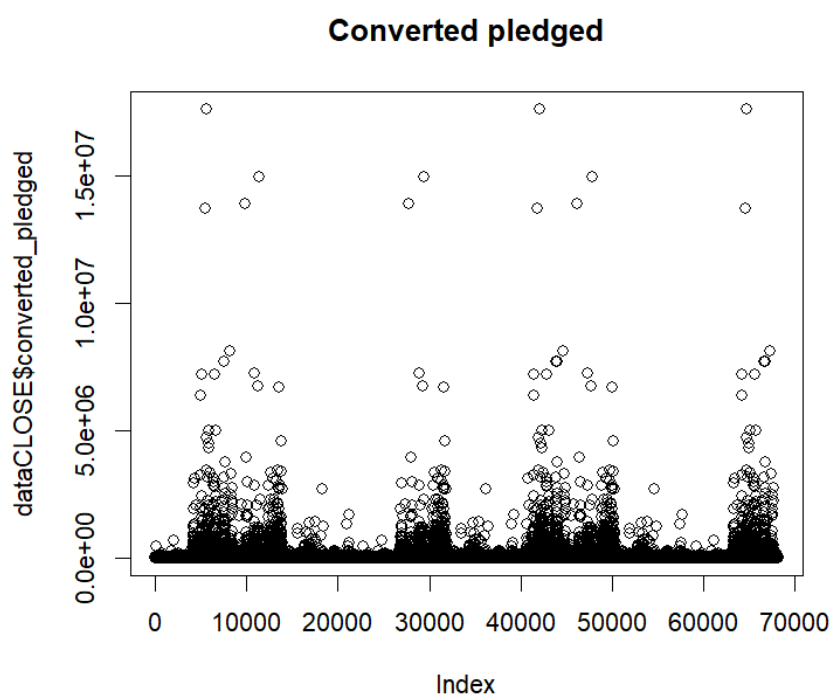


Fig. 11 – Converted pledged amount

(Source: own study)

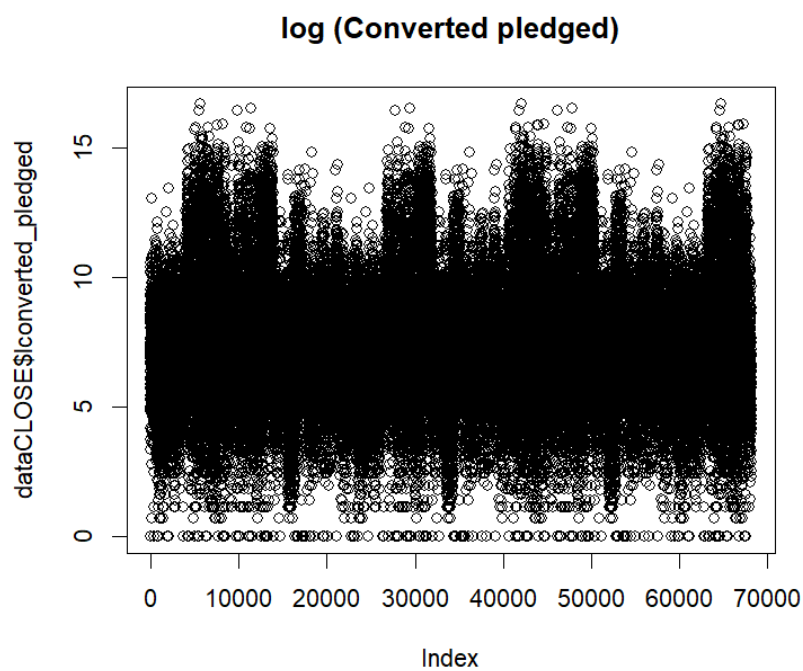


Fig. 12 – Log (Converted pledged amount)

(Source: own study)

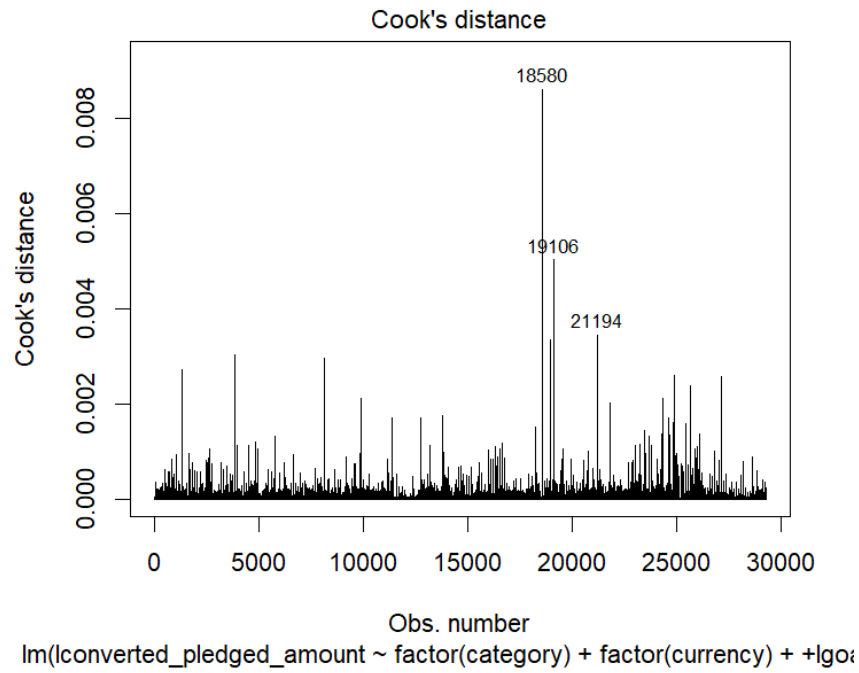


Fig. 13 – Outliers in the Kickstarter Linear model
(Source: own study)

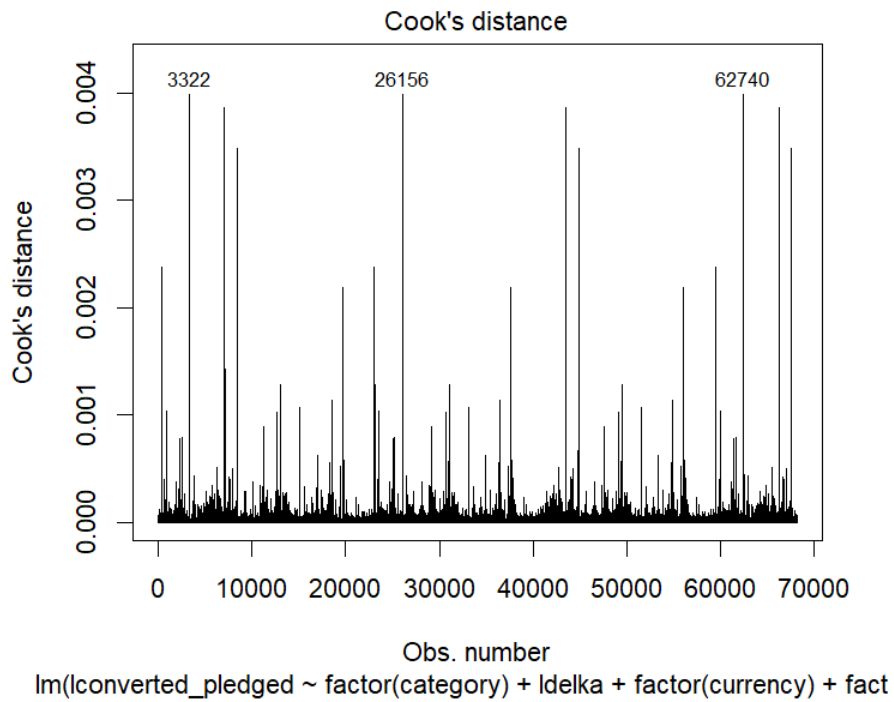
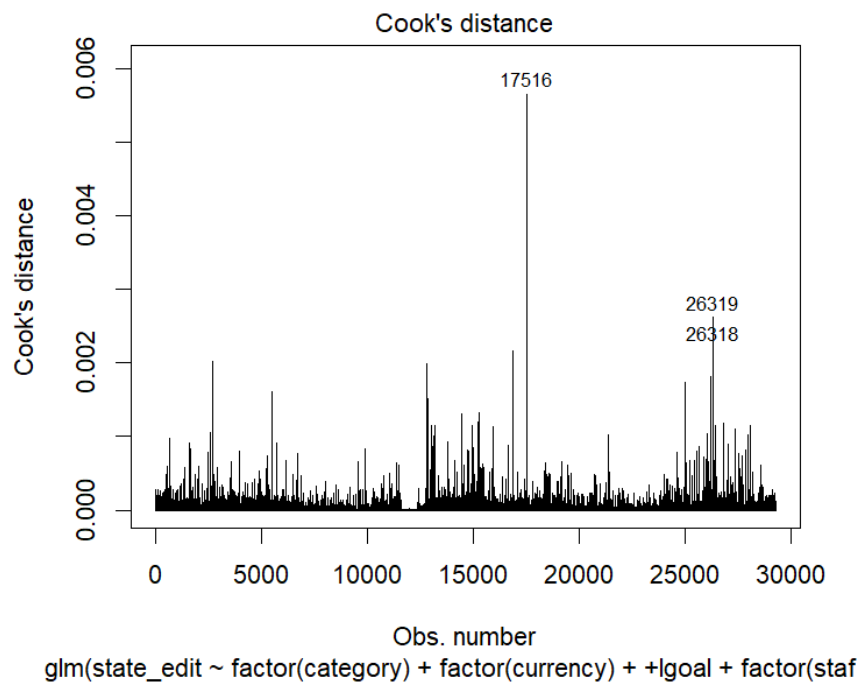
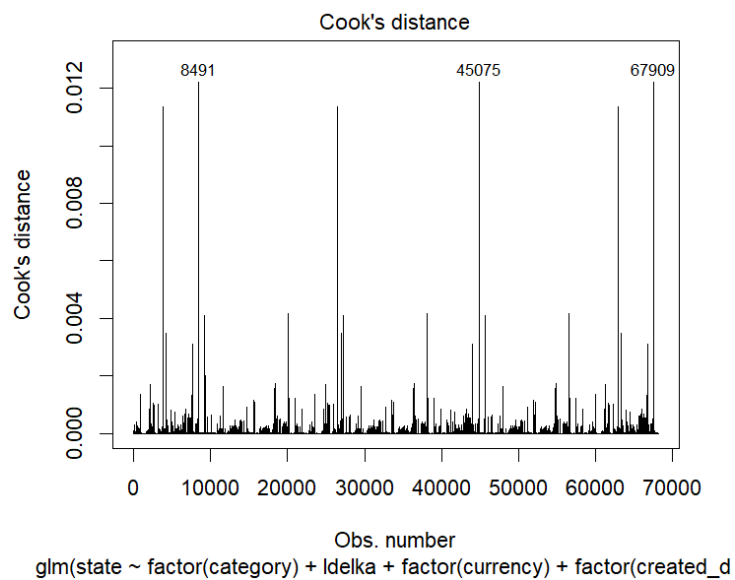


Fig. 14 – Outliers in the IndieGoGo Linear model
(Source: own study)



*Fig. 15 – Outliers in the Kickstarter Logistic model
(Source: own study)*



*Fig. 16 – Outliers in the IndieGoGo logistic model
(Source: own study)*

3.4.6 Multicollinearity test

Perfect multicollinearity has not been detected for the Kickstarter or IndieGoGo models. VIF factors for the models are present in the table below.

The requirement of no perfect multicollinearity is fulfilled for both dataset models. The vif factors for both final logistic models and models built on datasets with outliers are presented below.

	Model 1	Model 2
Category	2.6	3.2
Currency	1.4	1.2
Log (Goal)	1.4	1.1
Staff_pic	1.2	-
Log (Duration)	1.1	1.4
Created day	1	1
Created month	1.3	1.4
Created year	5.9	3.8
Spotlight	1.5	-
Starrable	2.6	-
Indemand	-	1

*Tab. 2 – VIF factors for the linear regression, Kickstarter and IndieGoGo
(Source: own study)*

3.4.7 Exogeneity test

Exogeneity assumption was tested visually, the relations between the residuals and each independent variable were depicted. No dependence was detected. Another good way to test the mean independence would be to plot the residuals against the number of rows.

Unfortunately, due to the large amount of data for this study, it was not possible to plot the residuals due to computational limitations. The exogeneity diagnostic plots are placed at the appendix section.

3.4.8 Normality of residuals test

The normality of residual distribution was verified by the Kolmogorov-Smirnov test. The null hypothesis was rejected with a p-value close to 0. However, due to the large sample of observation at the datasets the OLS estimator's distribution is asymptotically normal. Even though the assumption of normality of residuals is not met, standard regression test work correctly.

3.4.9 Linearity test

Logistic regression assumes linearity of independent variables and log odds for continuous variables. The graphs below show that for both models this requirement was not fully satisfied.

It may be worth using the generalized additive model (GAM) in the further studies.

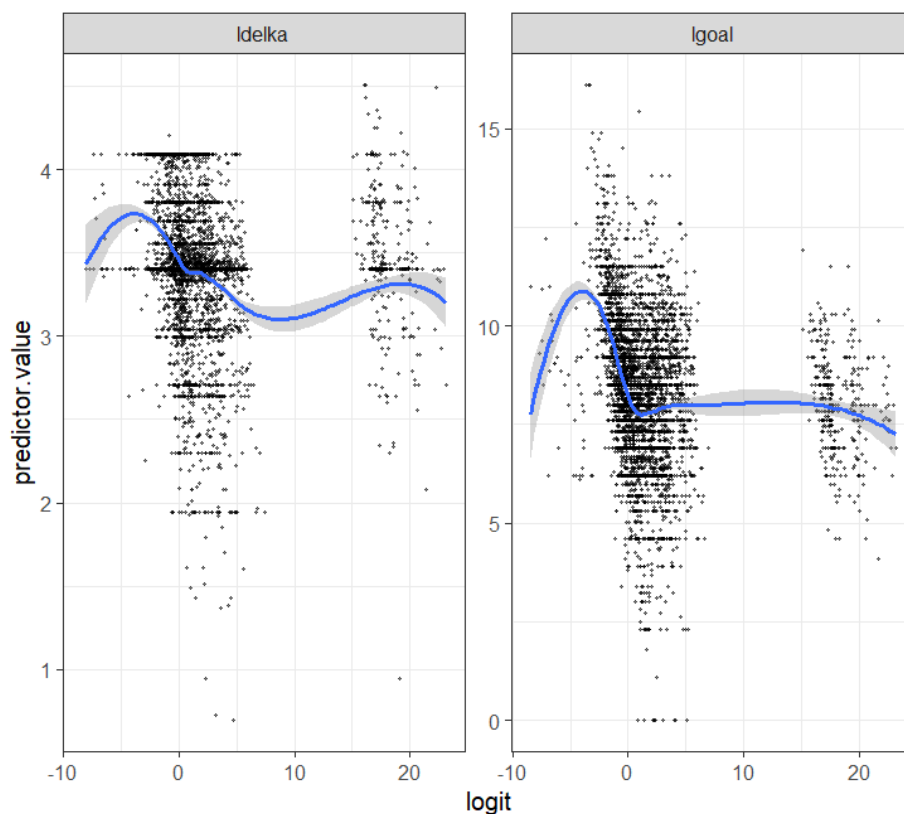


Fig. 17 – Relation between the independent variables and log-odds of succuess, Kickstarter
(Source: own study)

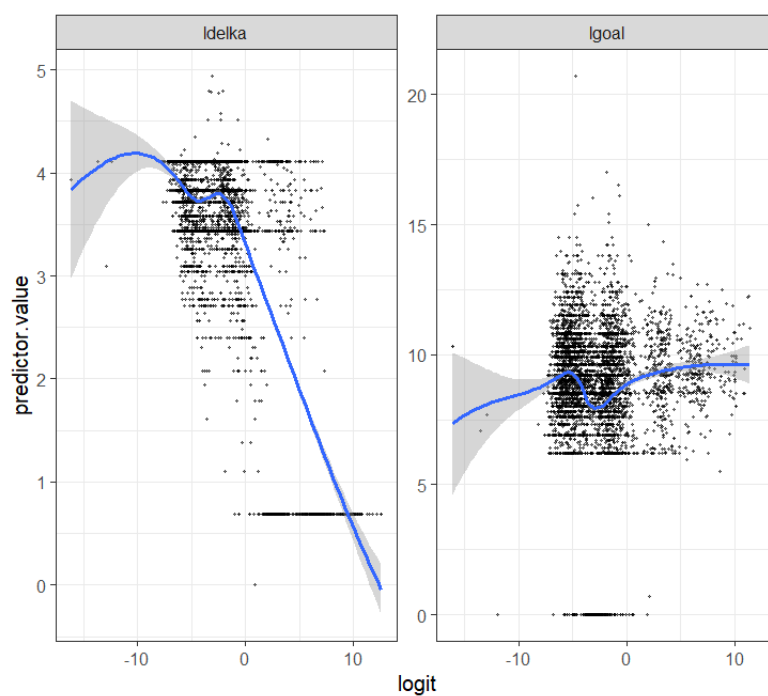


Fig. 18 – Relation between the independent variables and log-odds of succuess, IndieGoGo
(Source: own study)

	Q1	Q2	Q3	Q4
Kickstarter goal	7033	7175	7756	7322
Kickstarter duration	6816	1061	14074	7335
IndieGoGo goal	17031	15570	18492	17032
IndieGoGo duration	11769	21764	8097	26495

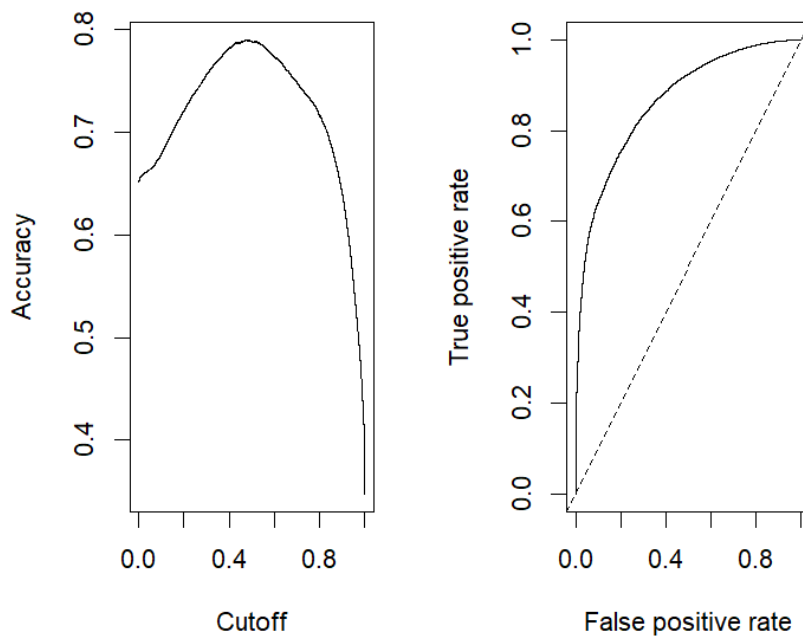
Tab. 3 – Quantile ranking of the independent variables Goal and Duration
(Source: own study)

3.5 ROC curve

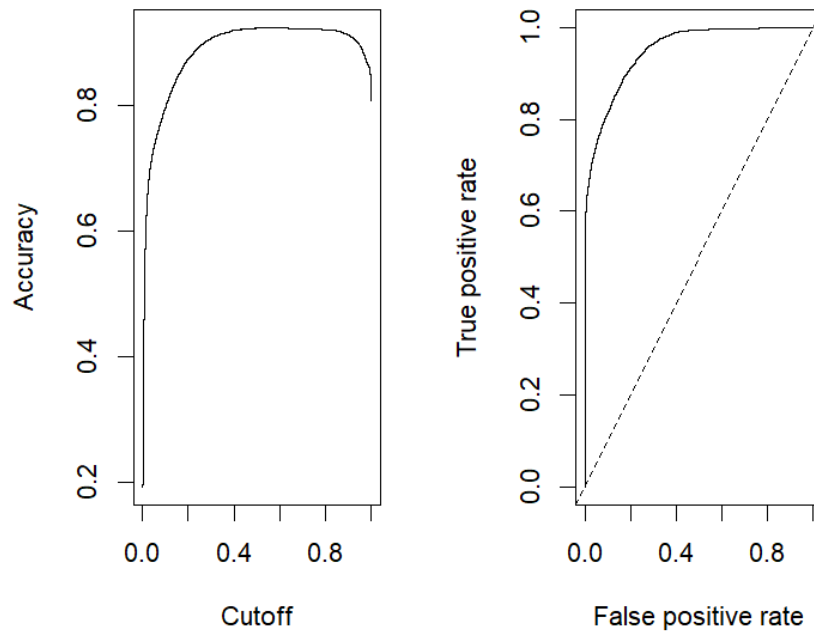
Mc Fadden pseudo R^2 for the Kickstarter model is 0.338, Log-Likelihood value is -12530 with 58 df. For the IndieGoGo model figures are 0.599 and -13396 with 74 df.

Another way of measuring the quality of the model is ROC curve (receiver operating characteristic curve).

The ROC is a curve generated by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings while the AUC is the area under the ROC curve. As a rule of thumb, a model with good predictive ability should have an AUC closer to 1 (1 is ideal) than to 0.5.



*Fig. 19 – ROC curve, Kickstarter Logistic model
(Source: own study)*



*Fig. 20 – ROC curve, IndieGoGo Logistic model
(Source: own study)*

3.6 Test and train datasets

Train test split is a model validation procedure that allows to simulate how a model would perform on new/unseen data. Splitting helps to avoid overfitting and to improve the training dataset accuracy.

13,000 and 4,000 observations were randomly selected from the original datasets for additional analysis model quality, making up the TRAIN and TEST datasets.

I set boundary at 0.5. If probability of success depends on variables in the model above will be higher then 50% then the project will be considered as successful one.

The classification prediction accuracy is about 76% for Kickstarter and 78% for the IndieGoGo models (misclassification error rates are 24% and 18%).

3.7 Prediction tables

One of the methods used to assess the quality of the model was the prediction table. The prediction table compares predictions of success or failure based on a regression model with actual data.

The first cell shows the number of truly successful projects that have been correctly identified as potential successes, the last cell shows the number of unsuccessful projects that have been correctly identified as failures.

The correct score was 79% for the Kickstarter model and 78% for the IndieGoGo.

Kickstarter Reality	Prediction	
	Success	Unsuccess
Success	6886	3309
Unsuccess	2894	16197

*Table 4 – Prediction table Kickstarter
(Source: own study)*

IndieGoGo Reality	Prediction	
	Success	Unsuccess
Success	54091	894
Unsuccess	4413	8727

*Table 4 – Prediction table IndieGoGo
(Source: own study)*

Discussion

1. The trend of the outliers in the models requires further investigation. It is difficult to establish the reasons for the additional trend, as there is insufficient information about the process of collecting the datasets used in this study. The best solution for the model fit might be the Quantile regression use.
2. In this study variable created year was used in the models as a factor, because of the large sample size datasets. In the studies with a smaller amount of the observation is recommended using the time variables as a trend before due to degree of freedom.
3. The assumption of linearity is not satisfied for the logistic regression model. In this study was used the logarithmic transformation of the continuous independent variables (goal and duration). The GAM method could be another option.

Conclusion

In this paper study the phenomenon of crowdfunding was discussed and reviewed through research on the success of projects. The linear and logistic regression A thorough diagnosis of the built models was made and possible further research directions were suggested. A table of the key factors influencing both the amount of money received and the probability of the projects success.

The study has shown that the day, the month and the year when the project was started could have an impact on its success. The best day to start a project is Tuesday, the worst is Saturday. The month when the project starts is of less importance. Projects started in July and December have the lowest chance of success. Probably because people spend more money than usual during the holiday season on gifts and traveling.

Since we analysed projects spanning more than 10 years, the regression coefficients obtained allow us to consider the impact of the covid. Covid had little influence on the success of the project and a positive influence on the amount of funds raised.

The regression results confirmed the high impact of the project category. Projects in the audio and board game categories have the highest chances of success. Individual countries were represented in the work by the currency variable. According to the results, projects launched in Japan collect the least. However, these projects have high chances of success.

The main achievement of the work is to develop the best possible models for the given analysis methods and to investigate the difficulties in processing this dataset for possible further research, as well as to compare 4 models with more variables in order to better understand the relationship between the dependent and independent variables.

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1. Appendix

Estimated coefficients 1

	Model 1	Model 2	Model 3	Model 4
Constant	1.635*** (0.19)	2.556*** (0.15)	22.181 (492.681)	-15.021 (113.03)
Comics	-0.29** (0.09)	-0.119** (0.04)	1.407*** (0.11)	2.091*** (0.25)
Camera Gear	—	1.105*** (0.06)	—	4.486*** (0.24)
Culture	—	0.089* (0.04)	—	3.469*** (0.24)
Education	—	-0.44*** (0.05)	—	-0.816* (0.41)
Green tech	—	-0.282*** (0.06)	—	3.419*** (0.24)
Environment	—	-0.507*** (0.05)	—	-0.123 (0.37)
Fashion	—	-0.603*** (0.04)	—	0.521 (0.3)
Health and fitness	—	-0.573*** (0.05)	—	0.953*** (0.28)
Home	—	-0.57*** (0.05)	—	1.43*** (0.27)
Human rights	—	0.38 (0.05)	—	3.49*** (0.24)
Local businesses	—	-0.606*** (0.05)	—	-0.246 (0.38)
Phones	—	-0.923*** (0.05)	—	0.584 (0.29)
Photography	—	0.237*** (0.05)	—	3.509*** (0.24)
Podcasts	—	-0.011 (0.05)	—	3.764*** (0.24)
Productivity	—	-0.233*** (0.05)	—	3.354*** (0.24)

Note: ***p < 0.01, **p < 0.01, *p < 0.05

(Source: own study)

Estimated coefficients 2

	Model 1	Model 2	Model 3	Model 4
Audio	—	1.327*** (0.06)	—	4.886*** (0.24)
Tabletop games	—	0.283*** (0.05)	—	4.492*** (0.24)
Transportation	—	0.249*** (0.06)	—	3.9*** (0.24)
Travel	—	-0.53*** (0.05)	—	0.996*** (0.28)
Video Games	—	-1.282*** (0.05)	—	-0.285 (0.36)
TV shows	—	-0.304*** (0.04)	—	-0.637 (0.34)
Wellness	—	0.365*** (0.05)	—	3.02*** (0.24)
Design	0.13 (0.91)	—	2.716*** (0.13)	—
Film	-0.362*** (0.1)	0.233*** (0.04)	0.222* (0.11)	1.26*** (0.26)
Food	-0.498*** (0.09)	-0.52*** (0.04)	0.58*** (0.11)	0.069 (0.3)
Games	0.645*** (0.1)	—	-0.259* (0.11)	—
Journalism	-0.8*** (0.1)	—	-0.184 (0.12)	—
Music	-0.488*** (0.9)	0.306*** (0.04)	0.431*** (0.11)	0.518 (0.28)
Publishing	-0.694*** (0.08)	-0.088* (0.04)	0.121*** (0.11)	1.194*** (0.27)
Technology	-0.285** (0.1)	—	-0.59*** (0.11)	—

Note: ***p < 0.01, **p < 0.01, *p < 0.05

(Source: own study)

Estimated coefficients 3

	Model 1	Model 2	Model 3	Model 4
Theater	-0.324*** (0.1)	0.432*** (0.04)	0.111*** (0.11)	0.305 (0.37)
CAD	0.027 (0.09)	0.504*** (0.06)	0.12 (0.12)	0.4* (0.16)
DKK	-0.997*** (0.18)	1.382*** (0.22)	0.6* (0.23)	-1.841* (0.71)
EUR	0.109 (0.86)	0.412*** (0.06)	-0.448*** (0.11)	0.136 (0.15)
GBP	0.394*** (0.08)	0.706*** (0.06)	0.263* (0.11)	0.446** (0.15)
HDK	-0.246* (0.12)	0.307** (0.1)	1.634*** (0.22)	1.461*** (0.2)
CHF	0.245* (0.25)	0.105 (0.1)	0.01 (0.28)	-1.894*** (0.53)
JPY	-2.16*** (0.15)	-1.49*** (0.16)	0.95*** (0.24)	0.555 (0.55)
MXN	-2.1*** (0.12)	-0.648* (0.29)	-0.326* (0.15)	-12.591 (132.72)
NOK	-1.332*** (0.26)	-0.385 (0.65)	1.047** (0.35)	1.837 (0.98)
NZD	0.338 (0.19)	-1.182** (0.38)	-0.146 (0.23)	-10.496 (119.5)
PLN	-0.451 (0.55)	-2.515*** (0.57)	-0.263 (0.01)	-1.818 (2.03)
SEK	-0.635*** (0.13)	-1.152*** (0.21)	1.409*** (0.19)	-1.714* (0.86)
SGD	0.178 (0.17))	0.178 (0.05)	0.358 (0.26)	0.459 (0.32)
USD	0.316*** (0.07)	0.631*** (0.05)	0.196* (0.1)	0.407** (0.15)
Log (Goal)	0.561*** (0.01)	0.617 (0)	-0.274 (0.01)	0.144*** (0.01)

Note: ***p < 0.01, **p < 0.01, *p < 0.05

(Source: own study)

Estimated coefficients 4

	Model 1	Model 2	Model 3	Model 4
Staff pic	0.868*** (0.02)	—	2.735 (0.07)	—
Log (Duration)	0.053 (0.03)	-0.702*** (0.01)	-0.747 (0.04)	-1.136* (0.02)
Monday	-0.007 (0.05)	0.11 (0.03)	0.125 (0.07)	0.219 (0.06)
Tuesday	0.113* (0.05)	0.33 (0.03)	0.24*** (0.07)	0.537 (0.06)
Wednesday	0 (0.05)	0.084 (0.03)	0.06 (0.07)	0.164 (0.07)
Thursday	0.009 (0.05)	0.036 (0.03)	0.046 (0.07)	0.01 (0.07)
Friday	-0.058 (0.05)	0.034 (0.03)	-0.015 (0.07)	0.113 (0.08)
Saturday	-0.12* (0.05)	-0.074 (0.03)	-0.051 (0.08)	-0.063 (0.09)
February	0.009 (0.05)	-0.028 (0.04)	0.083 (0.08)	-0.083 (0.09)
March	0.038 (0.05)	0.044 (0.04)	0.161* (0.08)	-0.146 (0.01)
April	0.038 (0.05)	0.144 (0.04)	-0.007 (0.08)	-0.043 (0.08)
May	0.659 (0.05)	0.194 (0.04)	-0.055 (0.08)	-0.003 (0.08)
June	0.142** (0.05)	0.175 (0.04)	-0.042 (0.08)	0.02 (0.08)
July	-0.074 (0.05)	0.041 (0.04)	-0.415*** (0.08)	-0.034 (0.08)
August	0.018 (0.05)	0.126 (0.04)	-0.162* (0.08)	-0.021 (0.08)

Note: ***p < 0.01, **p < 0.01, *p < 0.05

(Source: own study)

Estimated coefficients 5

	Model 1	Model 2	Model 3	Model 4
September	0.017 (0.05)	0.26 (0.04)	-0.018 (0.08)	0.058 (0.08)
October	0.092 (0.05)	0.191 (0.04)	-0.048 (0.08)	-0.001 (0.08)
November	0.073 (0.05)	0.227 (0.04)	-0.061 (0.08)	0.382*** (0.08)
December	0.01 (0.06)	-0.057 (0.04)	-0.69*** (0.09)	0.25** (0.09)
2010	0.187 (0.1)	—	-0.356 (545.496)	—
2011	0.364*** (0.09)	-0.205 (0.2)	-0.404 (528.24)	11.156 (113.03)
2012	0.422*** (0.1)	-0.27 (0.15)	-0.663 (514.102)	11.64 (113.03)
2013	0.447*** (0.09)	-0.26 (0.14)	-0.843 (509.474)	11.922 (113.03)
2014	0.316*** (0.1)	-0.2 (0.14)	-18.027 (492.681)	11.955 (113.03)
2015	0.045 (0.1)	-0.031 (0.13)	-18.196 (492.681)	11.587 (113.03)

Note: ***p < 0.01, **p < 0.01, *p < 0.05

(Source: own study)

Estimated coefficients 6

	Model 1	Model 2	Model 3	Model 4
2016	0.314** (0.1)	0.242 (0.13)	-17.971 (492.681)	11.999 (113.03)
2017	0.415*** (0.1)	0.64 (0.13)	-18.19 (492.681)	12.116 (113.03)
2018	0.592*** (0.1)	0.84 (0.13)	-17.623 (492.681)	12.238 (113.03)
2019	0.521*** (0.1)	0.91 (0.13)	-17.352 (492.681)	12.575 (113.03)
2020	0.653*** (0.1)	1.159 (0.13)	-17.606 (492.681)	12.664 (113.03)
2021	0.764*** (0.1)	0.991 (0.13)	-17.488 (492.681)	12.781 (113.03)
2022	1.045*** (0.27)	0.167 (0.14)	-24.332 (492.681)	13.75 (113.03)
Spotlight	4.88*** (0.03)	—	—	—
Starrable	4.882*** (0.2)	—	—	—
Indemand	—	2.402 (0.03)	—	6.841* (0.11)

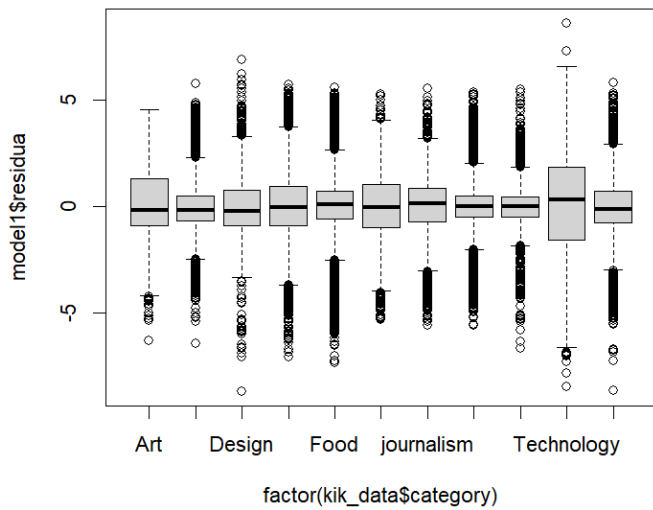
Note: ***p < 0.01, **p < 0.01, *p < 0.05

(Source: own study)

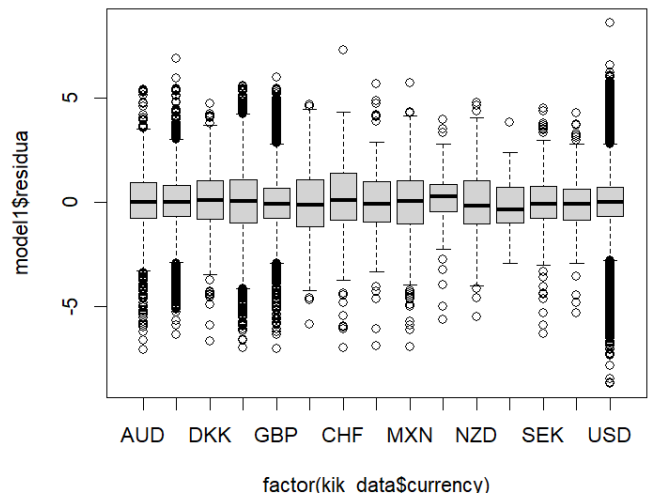
1.8 Exogeneity test

1.8.1 Exogeneity Kickstarter

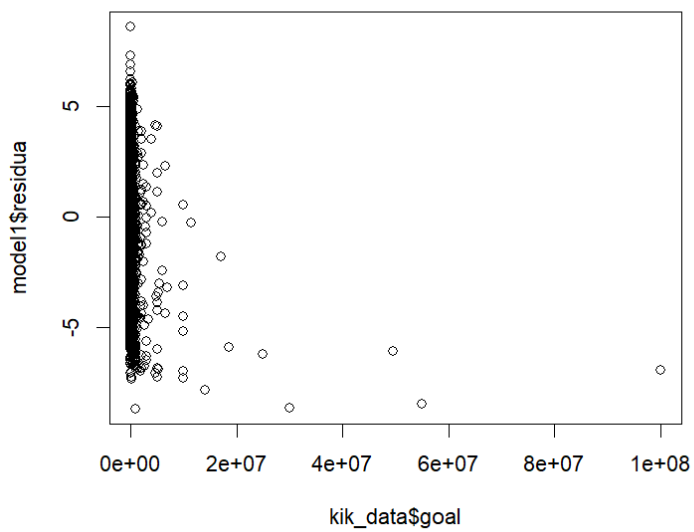
Residuals vs Category



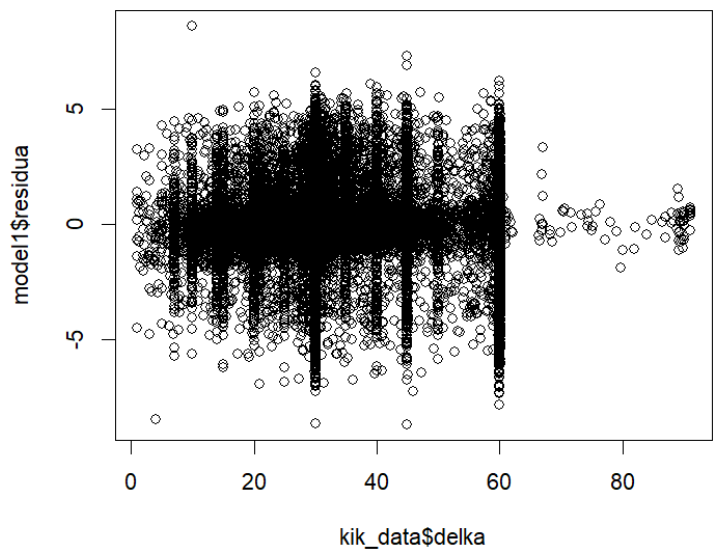
Residuals vs Currency

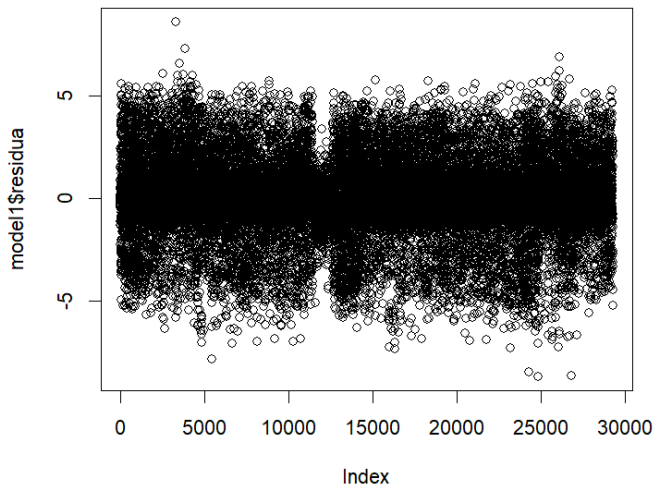
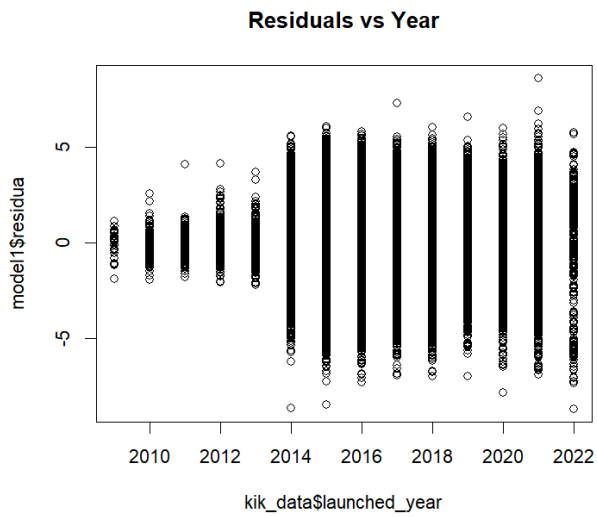
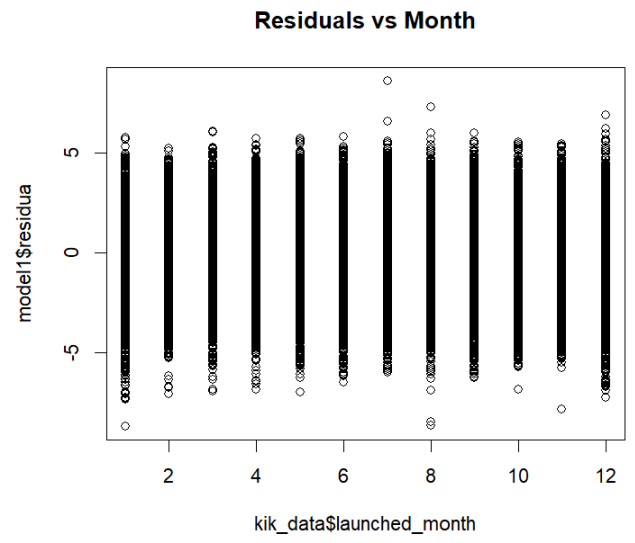
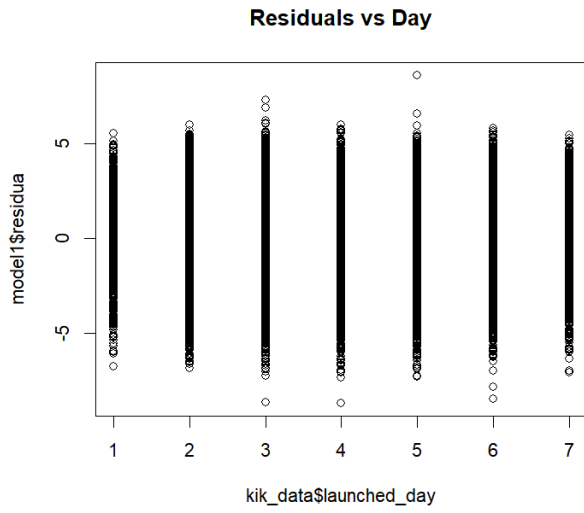


Residuals vs Goal

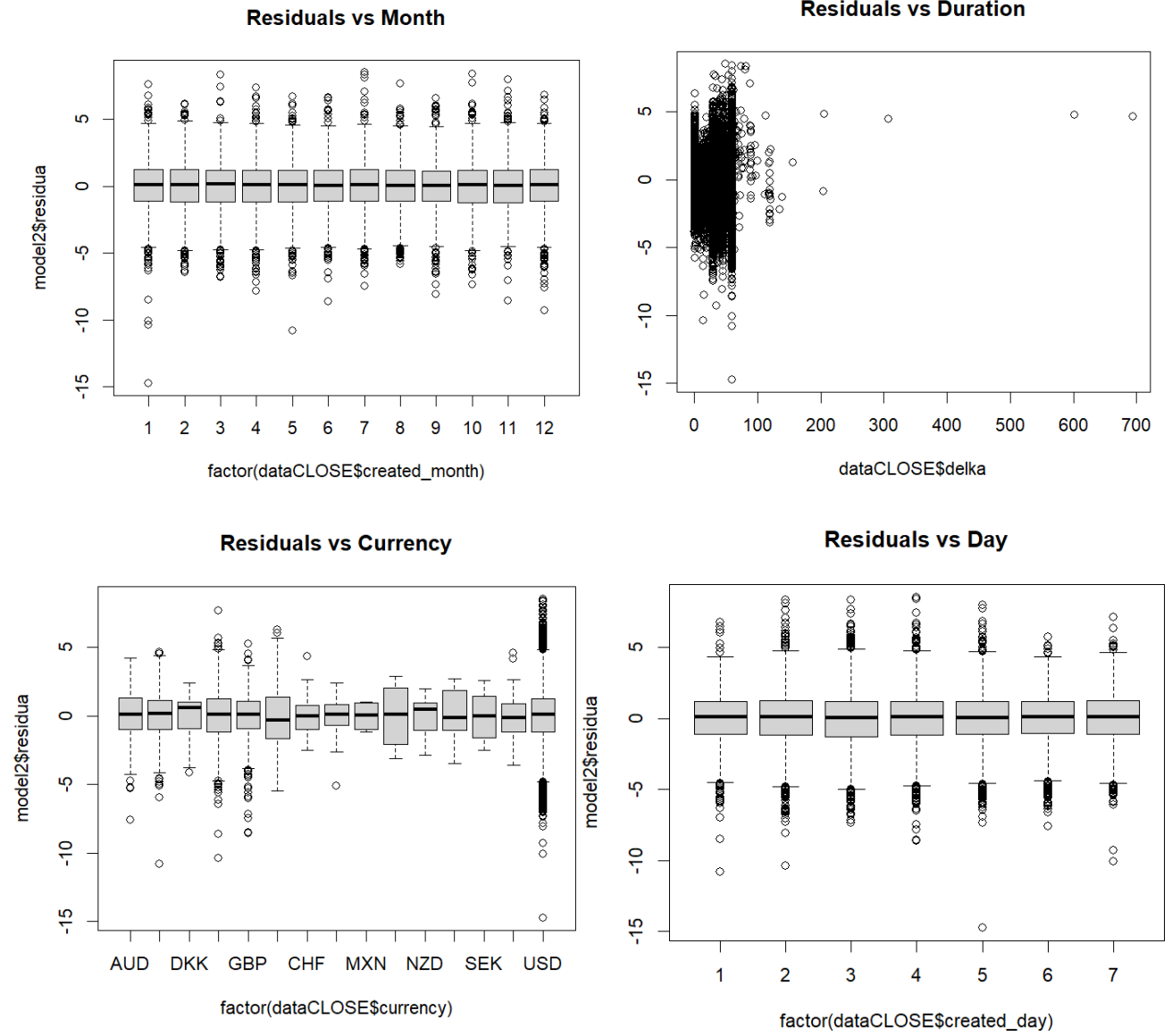


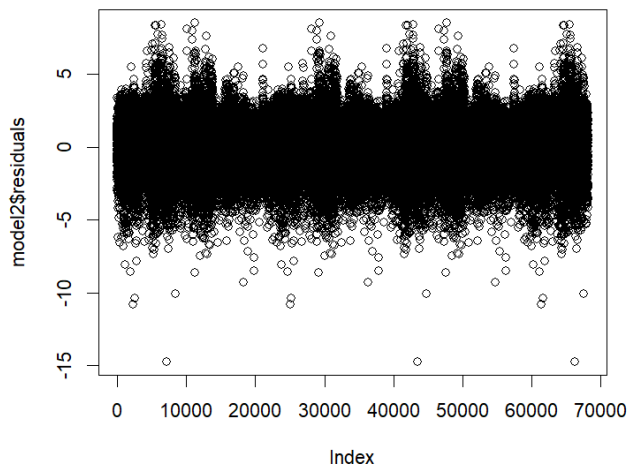
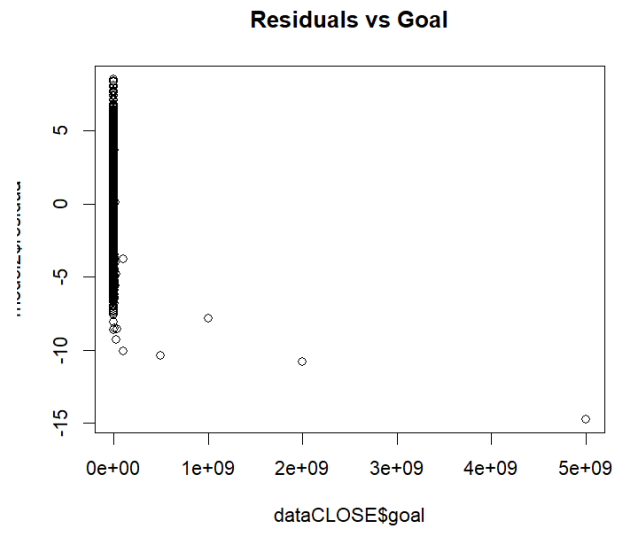
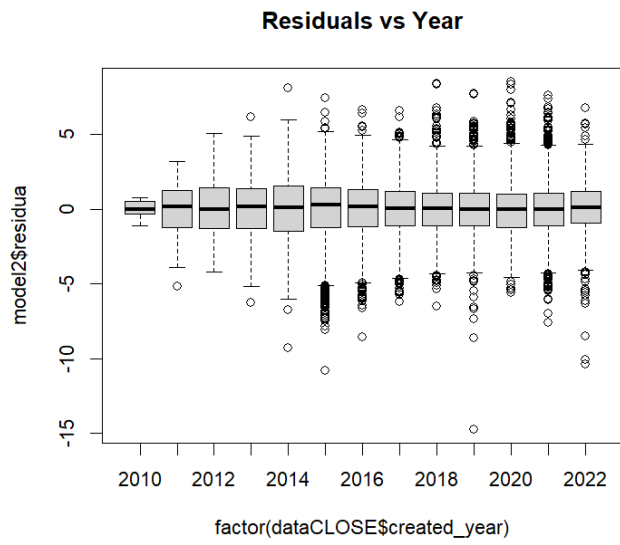
Residuals vs Duration



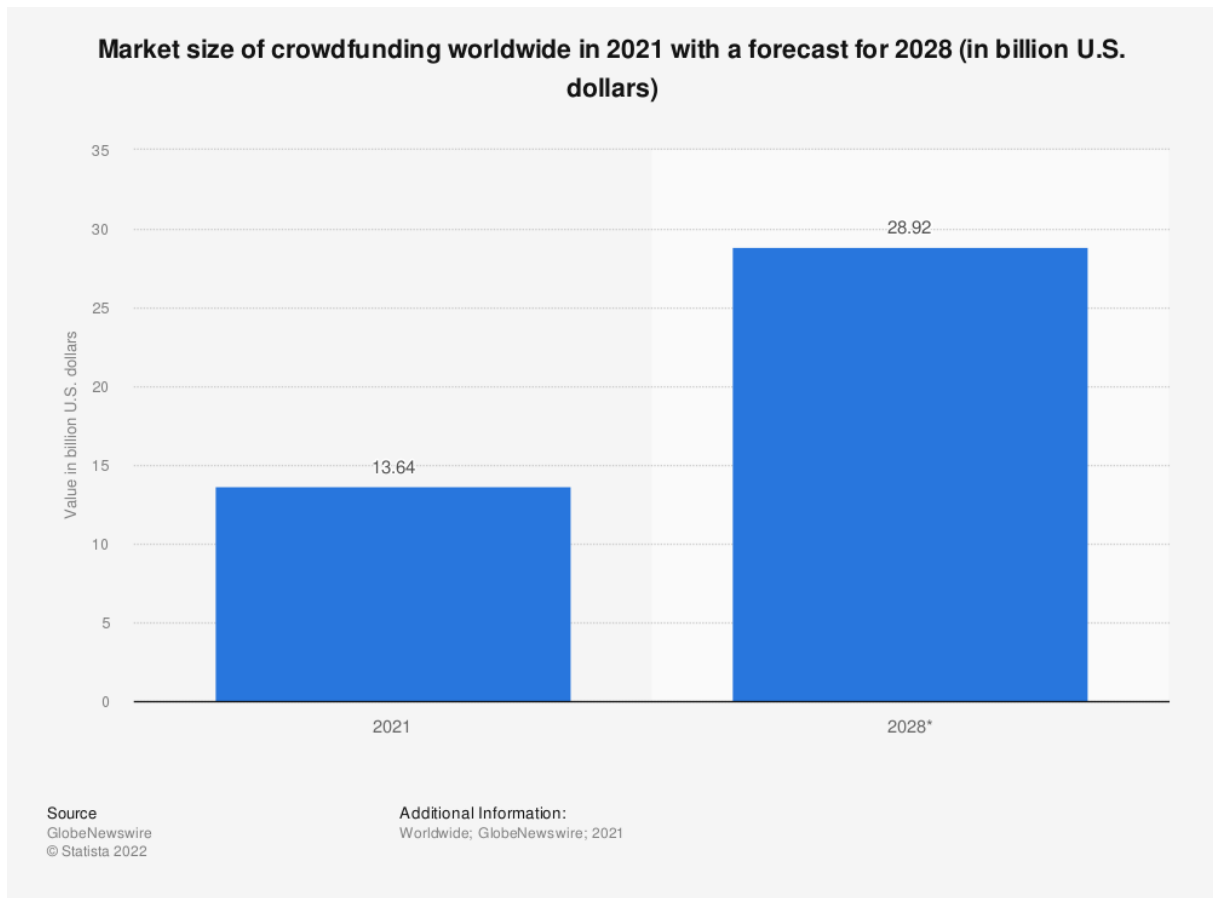


1.8.2 Exogeneity IndieGoGo



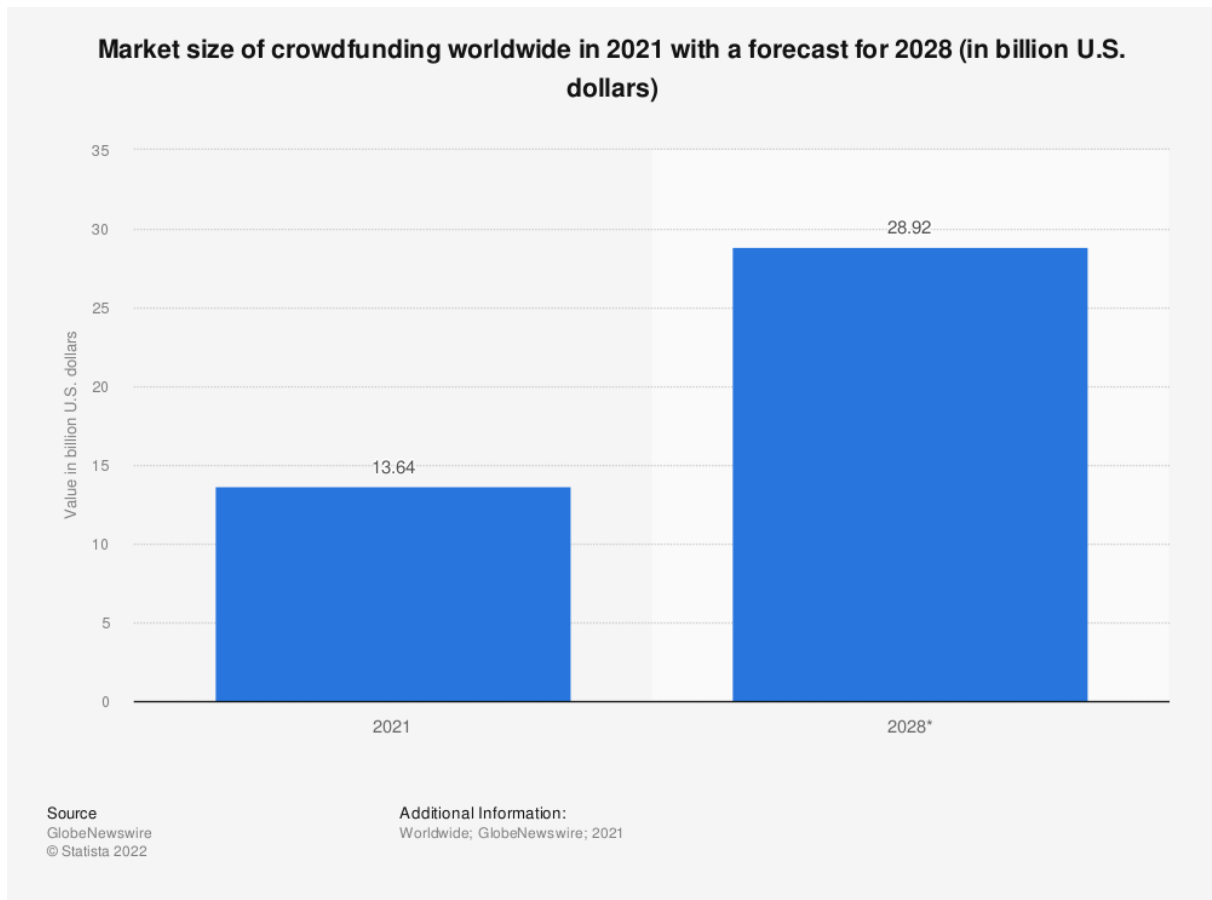


1.9 Market size of crowdfunding worldwide in 2021 with a forecast for 2028



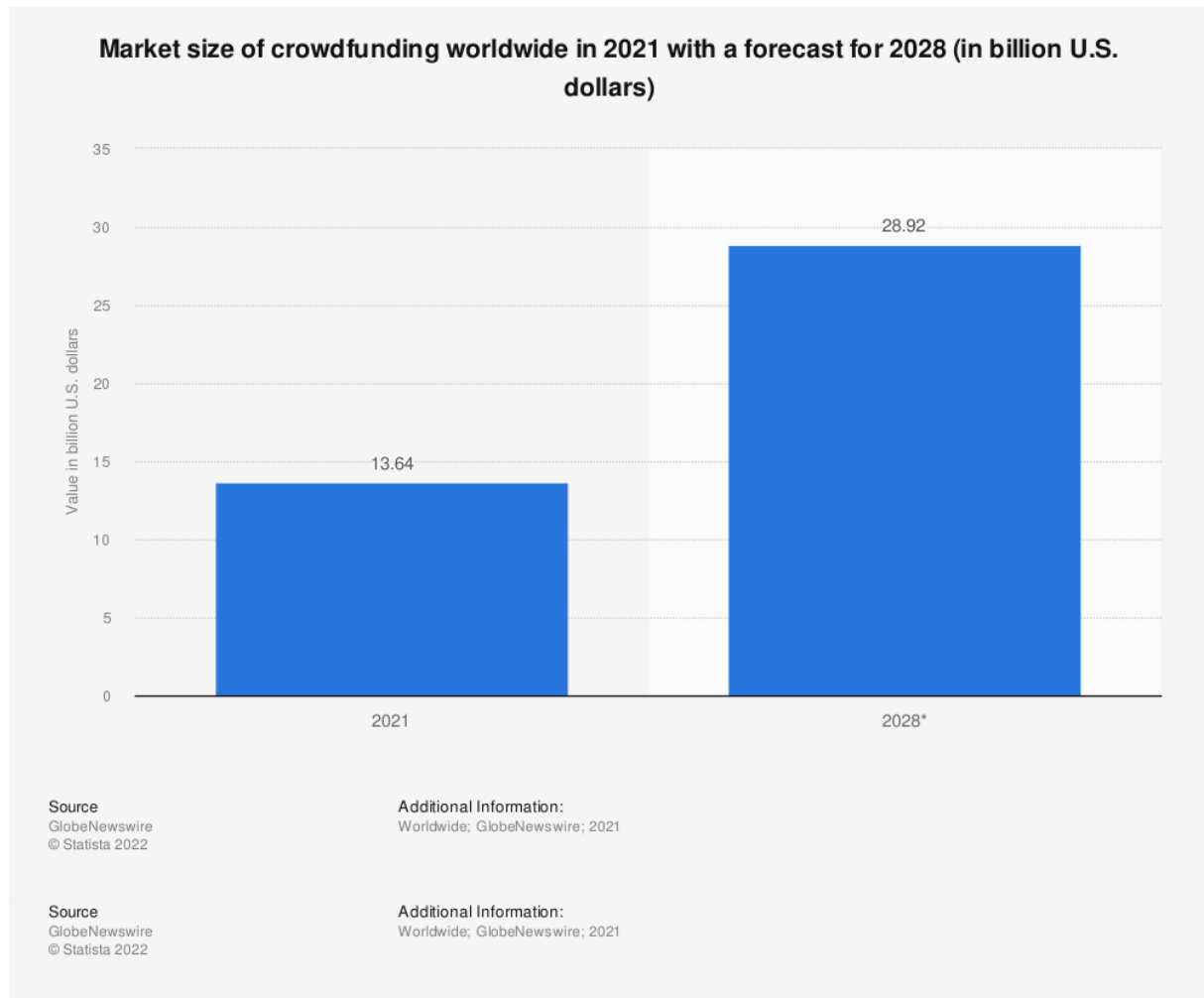
Statista. (2022). In 2021, the global crowdfunding market was valued at 13.64 billion U.S. dollars and was forecast to double by 2028, growing at compound annual growth rate (CAGR) of 11.2 percent. Available from: <https://www-statista-com.zdroje.vse.cz/statistics/1078273/global-crowdfunding-market-size/>

1.10 Market size of crowdfunding worldwide in 2021 with a forecast for 2028



Statista. (2022). In 2021, the global crowdfunding market was valued at 13.64 billion U.S. dollars and was forecast to double by 2028, growing at compound annual growth rate (CAGR) of 11.2 percent. Available from: <https://www-statista-com.zdroje.vse.cz/statistics/1078273/global-crowdfunding-market-size/>

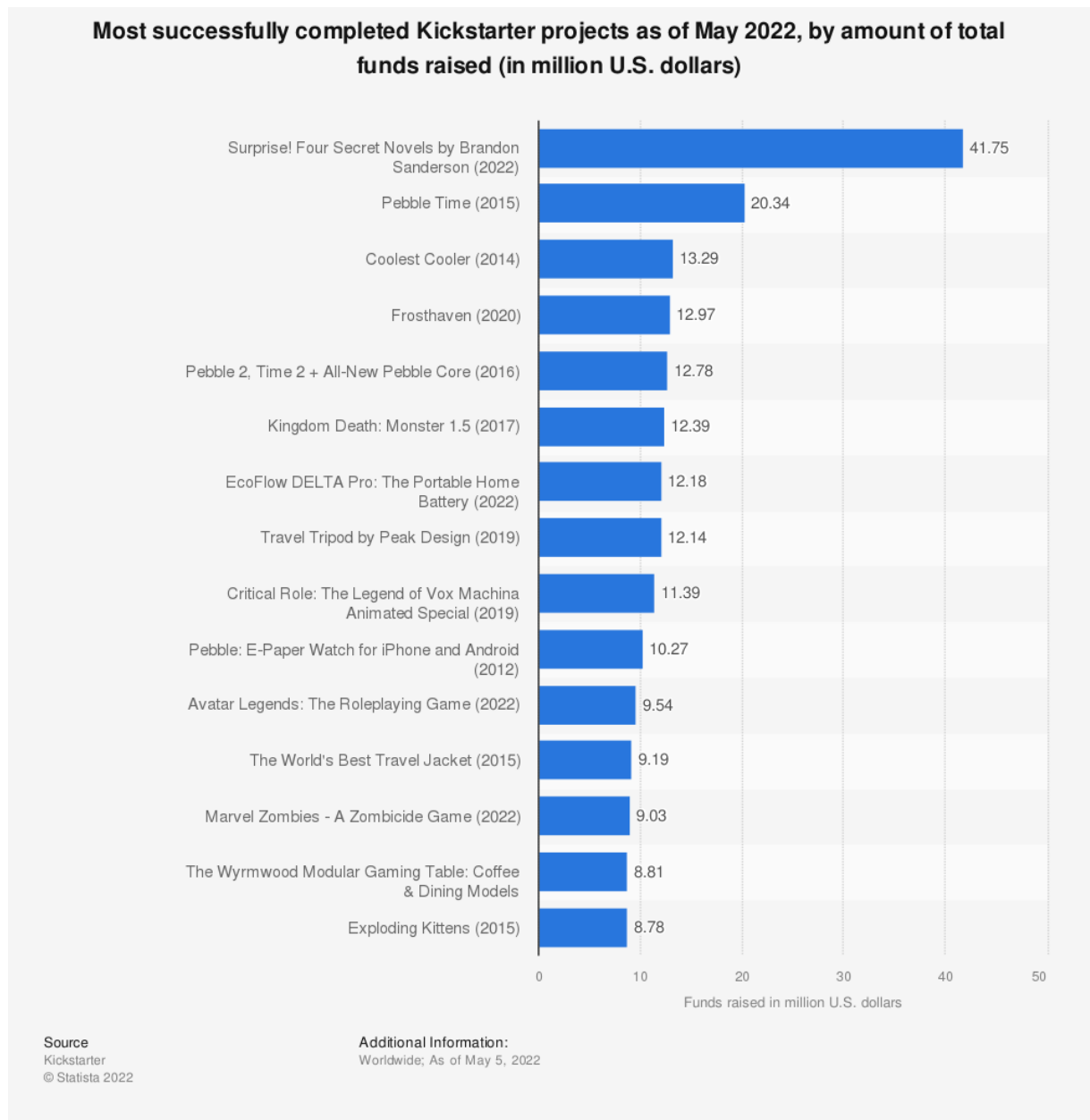
1.11 Market size of crowdfunding worldwide in 2021 with a forecast for 2028



1.12 Market size of crowdfunding worldwide in 2021 with a forecast for 2028

Statista. (2022). In 2021, the global crowdfunding market was valued at 13.64 billion U.S. dollars and was forecast to double by 2028, growing at compound annual growth rate (CAGR) of 11.2 percent. Available from: <https://www-statista-com.zdroje.vse.cz/statistics/1078273/global-crowdfunding-market-size/>

1.13 Most successfully completed Kickstarter projects as of May 2022, by amount of total funds raised



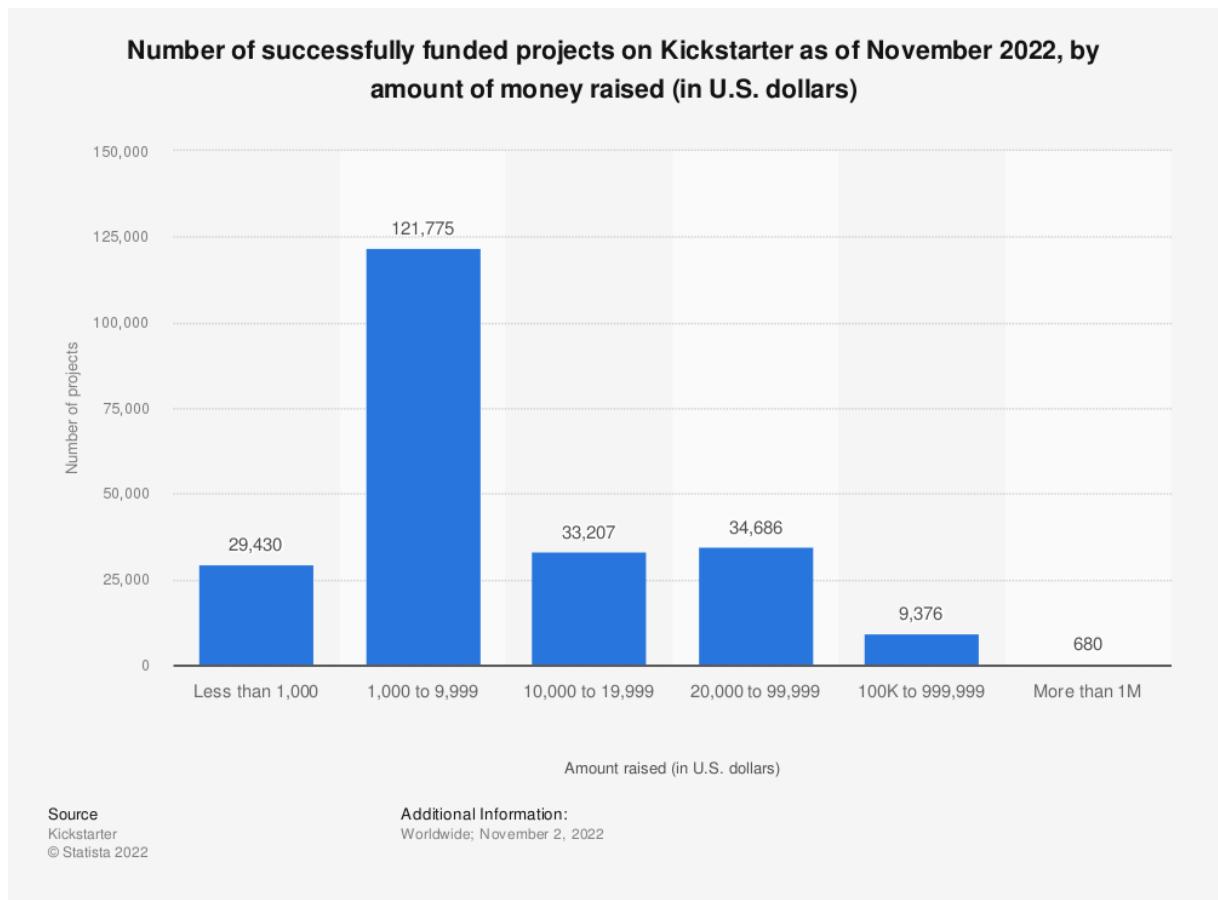
Statista. (2022, May 5). In 2022 the crowdfunding project Surprise! Four Secret Novels by Brandon Sanderson raised nearly 42 billion U.S. dollars, more than twice as much as Pebble Time. Available from: <https://www-statista-com.zdroje.vse.cz/statistics/222489/most-successful-completed-kickstarter-projects-by-total-funds-raised/>

1.14 Overview of projects and USD turnover on crowdfunding platform Kickstarter as of November 2022

Projects and dollars	Projects, million U.S. dollars, success rate in percent
Launched projects	574,262
Total dollars pledged (billion U.S. dollars)	6.94
Successful dollars (billion U.S. dollars)	6.33
Unsuccessful dollars (million U.S. dollars)	560
Live dollars (million U.S. dollars)	54
Live projects	3,181
Success rate (%)	40.13

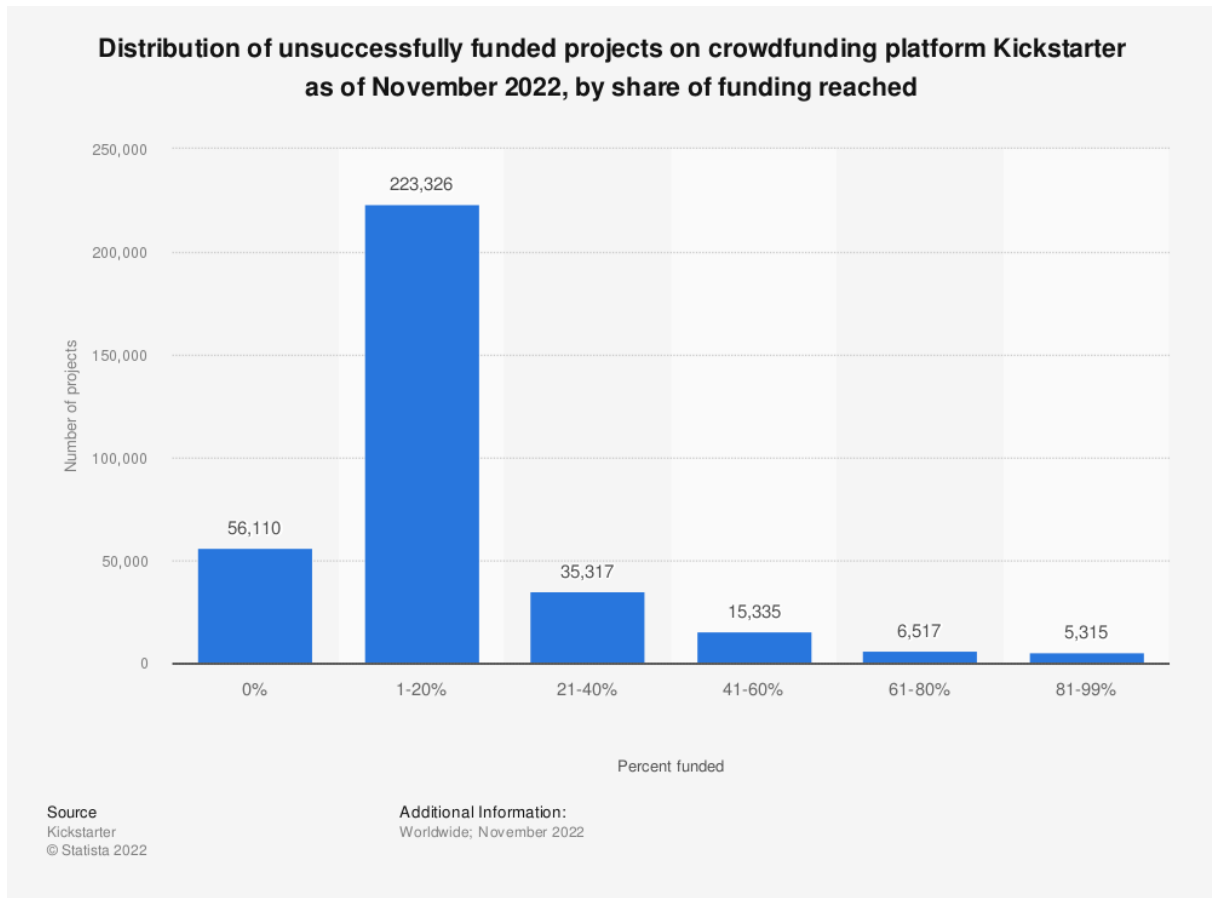
Statista. (2022, November). As of November 2022, the number of launched projects worldwide on the American crowdfunding platform Kickstarter amounted to 574,262. The success rate among these projects amounted to 40.13 percent. The total dollars pledged was 6.94 billion U.S. dollars, out of which 6.33 billion were successful dollars. Available from: <https://www-statista-com.zdroje.vse.cz/statistics/251727/projects-and-dollars-overview-on-crowdfunding-platform-kickstarter/>

1.15 Number of successfully funded projects on Kickstarter as of November 2022, by amount of money raised



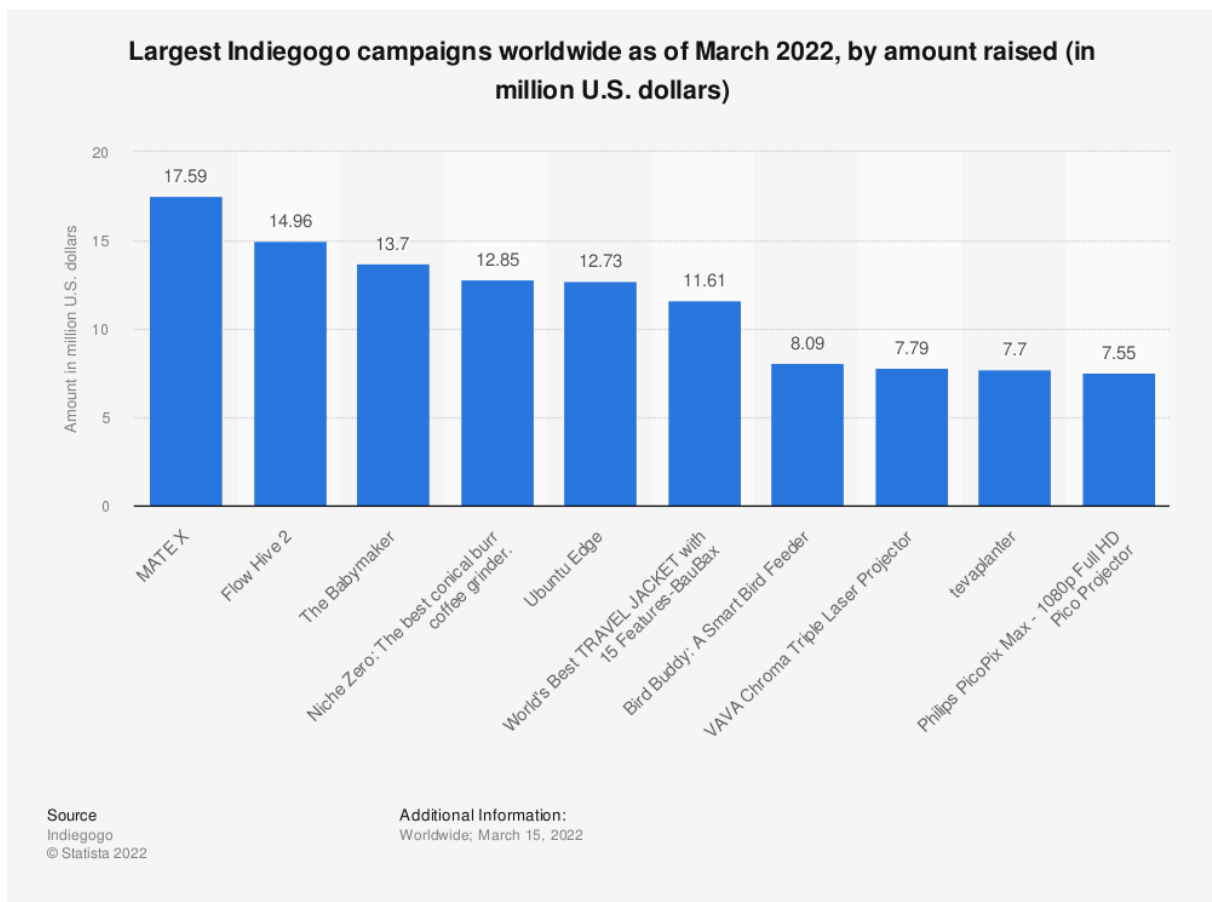
Statista. (2022, November 2). As of November 2022, most of the successfully funded projects on the crowdfunding platform Kickstarter raised between one and ten thousand U.S. dollars. On the other hand, the number of successfully funded projects at Kickstarter which raised more than one million U.S. dollars amounted to 680. Available from: <https://www-statista-com.zdroje.vse.cz/statistics/235397/distribution-of-kickstarter-funds-raised-by-successful-projects/>

1.16 Distribution of unsuccessfully funded projects on crowdfunding platform Kickstarter as of November 2022, by share of funding reached



Statista. (2022, November). The total number of unsuccessfully funded projects on the U.S. crowdfunding platform Kickstarter amounted to 341,922 up to November 2022. The number of projects that were 0 percent funded was 56,110 up to this point. Available from: <https://www-statista-com.zdroje.vse.cz/statistics/251732/overview-of-unsuccessfully-funded-projects-on-crowdfunding-platform-kickstarter/>

1.17 Largest IndieGoGo campaigns worldwide as of March 2022, by amount raised



Statista. (2022, March 15). The highest grossing IndieGoGo campaign as of March 2023 was for the MATE X. It is a foldable eBike, which raised nearly 17.6 million U.S. dollars on the platform. Available from: <https://www-statista-com.zdroje.vse.cz/statistics/620550/largest-IndieGoGo-campaigns-worldwide-by-amount-raised/>